

YOUTUBE CHANNEL: S KHAN ACADEMY

CS502 QUIZ FILE

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Subject Enrolment

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Final project (MTH620)

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1) The sieve technique works where we have to find ----- item(s) from a large input. **Single page 34**

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- 2) The only way to convert an empty string into a string of j characters is by doing j insertions, represented as **$E(0, j)=j$ page 78**
- 3) In Selection problem, the Sieve technique works in ----- **Phases page 34**
- 4) Algorithm is a sequence of computational steps that ----- the input into output. **Transform page 7**
- 5) In max heap (for Heap Sort algorithm), when every time maximum element is removed from top we replace it with ----- in the leaf tree. **Last page 41**
- 6) If p_j dominates p_i and p_i dominates p_h then p_j also dominates p_h . It means dominance relation is **Transitive page 18**
- 7) To find maximal points in brute-force algorithm, each point of space is compared against ----- of the space. **All other points page 13**
- 8) In the following code the statement “cout<<j;” executes ----- times
For($j=1; j \leq 5; j=j+2$)
Cout<<j; **3 times (conceptual)**
- 9) For average-case time analysis of Quick sort algorithm, Pivot selection is on average basis from -----.
All possible random values page 50
- 10) In the statement, “output $P[i].x, P[i].y$ ”, the number of times elements of P are accessed is ----- **2 page 14**
- 11) Identify the TRUE statement. **The Knapsack problem belongs to the domain of optimization problems Page 91**
- 12) Matrix multiplication is a(n) ----- operation. **Associative page 85**
- 13) ----- provides us more accurate result when input values are not closer to each other. **Median page 34(conceptual)**
- 14) In merge sort algorithm, we split the array around the ----- index q . **Mid page 28**
- 15) In the analysis of Selection algorithm, we get the convergent ----- series. **Geometric page 37**
- 16) In Selection problem, the Sieve technique ----- **Eliminate undesired data items each time page 35**

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YOUTUBE CHANNEL: S KHAN ACADEMY

- 17) If matrix A of dimension $p \times q$ is multiplied with matrix B of dimension $q \times r$, then each entry in resultant matrix takes ----- time. **$O(q)$ page 84**
- 18) In Sorting the key value or attribute ----- form an ordered domain. **Not always page 39**
- 19) In Heap Sort algorithm, the total running time for Heapify procedure is ----- . **Theta ($\log n$) page 43**
- 20) Pseudo code of algorithm are to be read by ----- . **People page 12**
- 21) In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object 'i', then we gain ----- . **V_i (value of object i)**
- 22) The worst-case running time of Quick sort algorithm ----- . **Is quadratic (google)**
- 23) The only way to convert a string of i characters into the empty string is with i deletions, represented as
• **$E(i, 0)=i$ page 78**
- 24) In Selection problem, the rank of an element will be its ----- position if we sort the input data. **Final page 34**
- 25) Which one is not passed as parameter in Quick sort algorithm? **Array (containing input elements) (cs301)**
- 26) While Sorting, the ordered domain means for any two input elements x and y ----- satisfies only. **All of the above page 39**
- 27) Which of the following is calculated with Big O notation? **Upper bounds page 25**
- 28) For solving Selection problem, we introduced Sieve technique due to ----- **Eliminating Rank of an element page 35**
- 29) The worst-case running time of merge sort is ----- in order to sort an array of n elements. **$O(n \log n)$ page 30**
- 30) Quick sort is a recursive algorithm. **True page 49**
- 31) While analyzing Selection algorithm, we make a number of passes, in fact it could be as many as ----- . **$\log(n)$ page 37**

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- 32) Approach for solving geometric problems by sweeping a line across the plane is called ----- sweep.
Plane page 18
- 33) ----- time is the maximum running time over all legal inputs. **Worst-case page 13**
- 34) The asymptotic growth of $n(n+1)/2$ is: **$O(n^2)$ page 15**
- 35) One of the limitations in 0/1 Knapsack is that an item can either be ----- in the bag or not. **Put page 91**
- 36) ----- is one of the few problems, where provable lower bounds exist on how fast we can sort. **Sorting page 39**
- 37) Selection algorithm takes theta ----- . **(n) page 37**
- 38) The Sieve technique is a special case, where the numbers of sub-problems is just ----- . **1 page 34**
- 39) Following is not the application of Edit Distance problem **Ascending Sort page 76**
- 40) We can use the optimal substructure property to devise a ----- formulation of the edit distance problem.
Recursive page 78
- 41) In plane sweep approach, a vertical line is swept across the 2d-plane form ----- . **Left to right page 18**
- 42) In 3-Dimensional space, a point P has ----- coordinate(s). **(X,Y,Z) (google)**
- 43) ----- items are not allowed in the 0/1 Knapsack. **Fractional page 91**
- 44) Radix sort is not a non-comparative integer sorting algorithm. **False (google)**
- 45) Fibonacci sequence was named on -----, a famous mathematician in 12th Century. **Leonardo Pisano page 73**
- 46) In generating Fibonacci Sequence, we can avoid unnecessary repetitions by ----- process.
Memorization page 74
- 47) The sequence of merge sort algorithm is: **Divide-Conquer-Combine page 27**
- 48) Sorting can be in ----- . **Both Increasing and Decreasing order page 52**

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YOUTUBE CHANNEL: S KHAN ACADEMY

49) The process of ----- ends when you are left with such tiny pieces remaining that it is trivial to solve them.

Divide and Conquer page 27

50) Time complexity of the “0-1” knapsack algorithm depends on ----- . **Number of items and capacity of knapsack (Google)**

51) In max-heap, largest element is stored at root node. Where is the smallest element stored? **Leaf Node (Google)**

52) For ----- values of n , any-----algorithm is fast enough. **Small pg#14**

53) To predict the cost of an algorithm in terms of resources is called ----- . **Analysis (Google)**

54) Dynamic programming comprises of ----- **Recursion only pg#75**

55) For Quick sort algorithm, Partitioning takes theta -----? **$n \log(n)$ (Google)**

56) An in-place sorting algorithm is one that ----- uses additional array for storage. **Does not pg#54**

57) In Dynamic Programming approach, we do not store the solution to each sub-problem in case if it reappears. **False pg#75**

58) Counting sort has time complexity: **$O(n+k)$ pg#58**

59) Due to left complete nature of binary tree, the heap can be stored in **Arrays pg#40**

60) For the heap sort we store the tree node in **Level-order traversal pg#40**

61) In the clique cover problem, for two vertices to be in the same group, they must be -----each other. **Adjacent to pg#176**

62) In in-place sorting algorithm is one that uses arrays for storage: **No additional array pg#54**

63) Brute-force algorithm for 2D-Maxima is operated by comparing ----- pairs of points. **All pg#18**

64) $F(n)$ and $g(n)$ are asymptotically equivalent. This means they have essentially the same ----- for large n . **Growth rates pg#23**

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- 65) Quick sort is **Not stable but in place** pg#54
- 66) Heaps can be sorted in arrays without using any pointers; this is due to the ----- nature of the binary tree.
Left-complete pg#40
- 67) Which may be a stable sort? **Both** above page 54
- 68) For the Sieve Technique we take time **$T(nk)$** page 34
- 69) One of clever aspects of heaps is that they can be stored in arrays without using any ----- . **Pointer's** page 40
- 70) Counting sort is suitable to sort the elements in range 1 to k. **K is small** page 57
- 71) Asymptotic growth rate of the function is taken over ----- case running time. **Worst** page 14
- 72) The Sieve technique is a special case, where the number of a sub-problems is just. **1** page 34
- 73) Dynamic programming algorithms need to store the results of intermediate sub-problems. **True** page 75
- 74) In counting sort, once we know the ranks, we simply ----- numbers to their final positions in an output array. **Copy** page 57
- 75) In Quick sort, we don't have control over the sizes of recursive calls. **True** page 49
- 76) Before sweeping a vertical line in a plane sweep approach, in start sorting of the points is done in increasing order of their ----- coordinates. **X** (page #18)
- 77) Random access machine or RAM is a/an **Mathematical model** (page #10)
- 78) The Huffman codes provide a method of encoding data inefficiently when coded using ASCII standard. **True** (page #99)
- 79) A heap is a left-complete binary tree that conforms to the ----- . **Heap order**
- 80) For the sieve technique we solve the problem, **recursively** (page #34)

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YOUTUBE CHANNEL: S KHAN ACADEMY

- 81) Median is not useful measure of central tendency of given input set especially when the distribution of values is highly skewed. **False (page # 34)**
- 82) The omega-notation allows us to state only the asymptotic-----bounds. **Lower (page #25)**
- 83) In plane sweep approach, a vertical line is swept across the 2-D plane and-----structure is used for holding the maximal points lying to the left of the sweep line. **Stack (page #18)**
- 84) When a heapify procedure is applied to the root node to restore the heap, then at each level, the comparison performed takes time. **It will take $O(1)$ (page #43)**
- 85) Efficient algorithm requires less computational----- . **Memory and running time (page #9)**
- 86) Recurrence can be described in terms of a tree. **Yes (page#31)**
- 87) Time complexity of dynamic programming-based algorithm for computing the minimum cost of Chain Matrix multiplication is----- . **n^3 (n cube) (page#90)**
- 88) In order to say anything meaningful about our algorithms, it will be important for us to settle on a----- . **Mathematical model of computation**
- 89) The iteration method is used for ----- . **Solving recurrence relation (page # 31)**
- 90) Chain matrix multiplication problem can be solved through-----strategy. **Dynamic programming (page #85)**
- 91) For comparison-based sorting algorithms, it is -----possible to sort more efficiently than Omega $n \log(n)$ time. **Not (page #54)**
- 92) Dynamic Programming approaches are usually useful in solving optimization problems. **True (page # 97)**
- 93) In sorting the key value or attribute-----from an ordered domain. **Must be (page #39)**
- 94) Result of asymptotical analysis of $n(n-3)$ and $4n*n$ is that----- . **both are asymptotically Equivalent (page #23)**
- 95) Floor and ceiling are-----to calculate while analyzing algorithms. **Usually consider difficult pg#31**
- 96) ----- of reference is an important fact of current processor technology. **Locality (page # 8)**
- 97) As per algorithm of dynamic programming, we need to store **Intermediate sub-problems (page #75)**

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YOUTUBE CHANNEL: S KHAN ACADEMY

- 98) In Sieve technique, we solve the problem----- **In recursive manner (page # 34)**
- 99) Upper bound requires that there exist positive constant c_2 and n_0 such that $f(n) \leq c_2 n$ for all $n \leq n_0$.
Equal to or less than (page # 24)
- 100) In 2-D matrix problem, A point p is said to be dominated by point q if----- **$P.x \leq q.x$ and $p.y \leq q.y$ (page # 11)**
- 101) A Random-Access Machine (RAM) is an idealized machine with----- random access. **Infinite large (page # 10)**
- 102) There are----- entries in the Edit Distance Matrix. **$\Theta(n^2)$ (page # 84)**
- 103) Counting sort is suitable for sorting the elements within range 1 to P , where----- **P is small (page # 57)**
- 104) While analyzing searching algorithm, we make a number of passes, in fact it called----- **$\log(n)$ (page #37)**
- 105) It is not a Fibonacci Sequence.
1,1,1,2,3,5,8,13,21,34,55, --- **True (page #73)**
- 106) In Random Access Machine (RAM), instructions are executed in----- **One by one (page # 10)**
- 107) Dynamic programming formulation of the matrix chain multiplication problem will store the solutions of each sub problem in $a(n)$: **Table (page #86)**
- 108) An algorithm is a mathematical entity, which is independent of----- **Programming language Compiler and Machine (page #7)**
- 109) In Fibonacci sequence, unnecessary repetitions do not exist at all. **True (page# 74)**
- 110) In Quick sort algorithm, pivots form----- **Binary Search Tree (page #49)**
- 111) In Heap Sort Algorithm, Heapify procedure is----- in nature. **Recursive (page #43)**
- 112) In Bucket sort, if there are duplicates then each bin can be replaced by a: **Link list (page #69)**
- 113) Counting sort assumes that the numbers to be sorted are in the range ----- **1 to k where k is small (page # 57)**
- 114) In Dynamic Programming, our approach is to ----- **Build the solution in a bottom- up fashion. Pg#92**

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YOUTUBE CHANNEL: S KHAN ACADEMY

- 115) In knapsack Problem, the goal is to put items in the knapsack such that the value of the items is -----
----- subject to weight limit of knapsack. **maximized**
- 116) If there are $\Theta(n^2)$ entries in edit distance matrix then each entry $E(i,j)$ takes ----- time to compute. •
 $\Theta(1)$ (page #84)
- 117) In ----- knapsack Problem, limitation is that an item can either be put in the bag or not.
Fractional items are not allowed. **0/1 (page #91)**
- 118) Heapsort is a/an ----- and ----- sorting algorithms. **In-place, not stable one (page#54)**
- 119) Selection sort is not an in-place sorting algorithm. **False (page #54)**
- 120) Memoization is a part of Dynamic Programming strategy. **True (page 74)**
- 27) Quick sort algorithms divide the entire array into-----sub arrays. **Two (Google)**
- 121) Insertion sort is an in-place algorithm. **True (page #54)**
- 122) When matrix A of 5×3 is multiply with matrix B of 3×4 then the number of multiplication required is: **60**
page 84
- 123) If matrix A of dimensions 2×4 is multiply with matrix B of dimensions 4×3 , then the dimensions of
resultant matrix is: **2×3 page 84(conceptual)**
- 124) Dynamic Programming strategy is useful when sub- problems are independent. **False google**
- 125) If Matrix-A of dimension 2×4 multiply with Matrix-B of dimension 4×3 , then the dimension of resultant
Matrix is: **2×3**
- 126) In the following code the statement “cout<< i” executes-----times.
for (int i=1; i<=n; i++)
cout<< i; **n times (conceptual)**
- 127) Edit distance algorithm based on-----strategy. **Dynamic programming page 77**

YOUTUBE CHANNEL: S KHAN ACADEMY

- 128) If Matrix A has dimensions “3x2” and Matrix B has dimensions “2x3”, then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions. **3x3 page 84(conceptual)**
- 129) The function $f(n) = n(\log n + 1)/2$ is asymptotically equivalent to $n \log n$. Here Lower Bound means function $f(n)$ grows asymptotically at ----- as fast as $n \log n$. **Least page 24**
- 130) Single atom from a larger set of -----, **n items page 34**
- 131) Fibonacci sequence was named on a-----, a famous mathematician in 12th century. **Leonardo Pisano (page # 73)**
- 132) In dynamic programming, our approach is to-----, **Build the solution in a bottom up fashion (page #92)**
- 133) In heap sort algorithm, we build-----for ascending order. **Max heap page 42(conceptual)**
- 134) Bubble sort is not an in-place sorting algorithm. **False (page #54)**
- 135) In partition algorithm, the subarray-----has elements which are greater than pivot element X. **A[q+1.....r] (page #46)**
- 136) In Heap sort algorithm, if heap property is violated-----, **We call heapify procedure (page # 41)**
- 137) -----is not a characteristic of Random-Access Machine. **Locality of reference (page #10)**
- 138) Boolean operation is a-----operation on an idealized RAM model of computation. **Basic (page #10)**
- 139) How much time merge sort takes for an array of numbers? **T(n log n) page 40**
- 141) Sorting is one of the-----problems in Computer Sciences. **Easiest page 39**
- 142) We can make-----recursive calls in Fibonacci sequence. **Infinite google**
- 140) Insertion sort is a ----- sorting algorithm. **In-place page 54**
- 141) We do not need to prove comparison- based sorting algorithms by mathematically, it always takes-----: **Omega nlog(n) page 54**

YOUTUBE CHANNEL: S KHAN ACADEMY

- 142) A Principal operation for maintaining the heap property is called heapify, it is also called: **Shifting Down page 43**
- 143) The number of nodes in a complete binary tree of height h is: **$2^{(h+1)}$ page 45**
- 144) After partitioning array in Quick sort, pivot is placed in a position such that: **Values smaller than pivot are on left and larger than pivot is on right page 35**
- 145) ----- programming is essentially recursion without repetition: **Dynamic page 75**
- 146) We can use the _____ property to devise a recursive formulation of the edit distance problem. **optimal substructure page 78**
- 147) Applying the sieve technique to selection problem, _____ element is picked from array. **Pivot page 35**
- 148) If matrix A of dimension $p \times q$ is multiplied with matrix B of dimension $q \times r$, then the dimension of the resultant matrix is: **$p \times r$ page 84**
- 149) In knapsack problem, each item must be entirely accepted or rejected, is called _____ problem. **“0-1” page 92**
- 150) In the case of _____ analysis does not depend upon on the distribution of input. **Quick sort page 50**
- 151) In Heapsort Algorithm, total time taken by heapify procedure is: **$O(\log n)$ page 43**
- 152) In merge sort algorithm, we split the array _____ to find index q . **midway page 28**
- 153) Al-Khwarizmi was a/an _____ **mathematician page 7**
- 154) In Sieve Technique, we know the item of interest. **False page 34**
- 155) Radix sort is a _____ integer sorting algorithm. **Non-comparative googles**
- 156) Random Access Machine (RAM) can execute _____ instruction. **logical and arithmetic page 10**
- 157) Which of the following is calculated with Big Omega notation? **Upper bounds page 25**
- 158) Asymptotic growth of $8n^2 + 2n - 3$ is: **$O(n^2)$ page 23**

YOUTUBE CHANNEL: S KHAN ACADEMY

159) _____ is the process of avoiding unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later. **Memorization page 74**

160) In Heap Sort algorithm, the first step is to _____. **Call Build-Heap procedure page 43**

161) The time assumed for each basic operation to execute on RAM model of computation is _____.
Constant page 10

162) If input “n” is odd, then median will be _____. **(n+1)/2 page 34**

163) In quick sort algorithm, _____ decides nature of Binary Search Tree formed by pivots. **rank of the pivot page 49**

164) The brute-force algorithm for 2D-Maxima runs in order $O(\text{-----})$ time: **$n*n$ page 18**

165) In average-case time analysis of quick sort algorithm, the most balanced case for partition is when we divide the list of elements into----- **Two nearly equal pieces google**

166) The main purpose of mathematical analysis is measuring the ----- required by the algorithm:
Execution time page 13

167) Dynamic Programming algorithms often use some kind of ----- to store the results of intermediate Sub-problems: **Table page 75**

168) The average case running time of quick sort algorithm is theta -----: **$n \log(n)$ page 53**

169) What is the best-case time complexity of merge sort? **$O((n \log n))$ page 40**

170) While solving Selection problem, in Sieve technique we partition input data -----: **According to Pivot page 35**

171) In average-case time the probability of seeing input is denoted by----- **$P(I)$ page 13**

172) To check whether a function grows faster or slower than the other function, we use some asymptotic notations, which is -----: **All of the given google**

173) Asymptotic notations are used to describe ----- of an algorithm: **Running time (google)**

174) In recursive formulation of Knapsack Problem
 $V[0,j] = \text{-----}$ for $j \geq 0$ **0 (page #93)**

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YOUTUBE CHANNEL: S KHAN ACADEMY

- 175) If a matrix has three rows and two columns, then dimensions of matrix will be: **3x2 (google0)**
- 176) Selection sort takes theta --- in the worst case: **(n²)**
- 177) Which of the following problems can be solved using dynamic problem? **Matrix chain multiplication problem page 92**
- 178) ----- algorithm based on Dynamic Programming strategy: **Edit distance page 77**
- 179) The definition of theta-notation relies on proving ----- asymptotic bound: **Both lower and upper page 25**
- 180) In order to say anything meaningful about our algorithms, it will be important for us to settle on a -----: **Mathematical model of computation page 10**
- 181) In asymptotical analysis of $n(n-3)$ and $4n*n$, as n becomes large, the dominant (fastest growing) term is some Constant times-----, **$n*n$ page 23**
- 182) ----- is in- place sorting algorithm **Bubble sort page 54**
- 183) In chain matrix multiplication, if there are ----- ways in which outer most pair of parentheses can placed: **$n-1$ page 85**
- 184) In chain matrix multiplication, table is filled ----- to find the multiplication of matrix: **diagonally page 86**
- 185) Which one sorting algorithm is best suited to sort an array of 2 million elements? **Quick sort google**
- 186) The problem with the brute-force algorithm is that it uses ----- in pruning out decisions: **No intelligence page 18**
- 187) Quicksort is a/an ----- and ----- sorting algorithm: **In-place , not stable one page 54**
- 191) In Heap Sort algorithm, the maximum levels an element can move upward is -----: **Theta (log n) page 43**
- 192) If the time complexity of an algorithm is $O(n)$, then it is called ----- time complexity: **Linear page 37**

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YOUTUBE CHANNEL: S KHAN ACADEMY

193) Dynamic Programming approach is usually useful in solving ----- problems: **Optimization**
page 97

194) What time does Merge Sort algorithm take in order to sort an array of 'n' numbers? **$\Theta(n \log n)$ page 30**

1. The sequence of merge sort algorithm is:

- a. Divide Combine-Conquer
- b. Conquer-Divide-Combine
- c. Divide-Conquer-Combine Page 27**
- d. Combine-Divide-Conquer

2. In Knapsack Problem, limitation is that an item can either be put in the bag or not. Fractional items are not allowed.

- a. 0
- b. 1
- c. 0/1 Page 91**
- d. Fractional

3. In Selection algorithm, we assume pivot selection takes theta running time.

- a. n**
- b. n^2
- c. n^3
- d. $\log(n)$

4. In Heap Sort algorithm (using max heap), when every time maximum elements removed from top .

- a. We call merge Sort Algorithm
- b. it becomes Order n^2 Algorithm
- c. Divide and Conquer strategy helps us
- d. We are left with a hole Page – 41**

5. If matrix A of dimension $p \times q$ is multiply with matrix B of dimension $q \times r$, then each entry in resultant matrix takes time.

- a. $O(q)$**
- b. $O(1)$
- c. $O(p \times q)$
- d. $O(q \times r)$

6. _ is a method of solving a problem in which we check all possible solutions to the problem to find the solution we need.

- a. Plane-Sweep Algorithm

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- ★ b. Sorting Algorithm
- ★ **c. Brute-Force Algorithm google**
- ★ d. Greedy approach
- ★ 7. The worst-case running time of Quick sort algorithm _.
- ★ a. Cannot be quadratic
- ★ **b. Is quadratic google**
- ★ c. Is always Exponential
- ★ d. Is linear
- ★ 8. In max heap (for Heap Sort algorithm), when every time maximum element is removed from top, we replace it with leaf in the tree.
- ★ a. second last
- ★ **b. Last Page -41**
- ★ c. First
- ★ d. Any
- ★ 9. Quick sort algorithm was developed by -
- ★ a. AlferdAho
- ★ b. Sedgewick
- ★ c. John Vincent Atanasoff
- ★ **d. Tony Hoare – Google wikipedia**
- ★ 10. If Matrix-A has dimensions "3x2" and Matrix-B has dimensions "2x3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.
- ★ a. 3x2
- ★ b. 2x3
- ★ c. 2x2
- ★ **d. 3x3**
- ★ 11. For comparison-based sorting algorithms, it is possible to sort more efficiently than Omega $n \log(n)$ time.
- ★ a. Always
- ★ **b. Not P-54**
- ★ c. Sometimes
- ★ d. Sometimes not
- ★ 12. Dynamic Programming approach is usually useful in solving optimization problems.
- ★ a. True
- ★ b. False
- ★ 13. In Sorting the key value or attribute from an ordered domain.
- ★ **a. Must be page 39**

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YOUTUBE CHANNEL: S KHAN ACADEMY

- ★ b. Not always
- ★ c. May be
- ★ d. Occasionally

★ 14. Result of asymptotical analysis of $n(n-3)$ and $4n*n$ is that _

- ★ a. $n(n-1)$ is asymptotically Less
- ★ b. $n(n-1)$ is asymptotically Greater
- ★ c. Both are asymptotically Not equivalent
- ★ d. Both are asymptotically Equivalent page 23 ($4n*n = 4n^2$)

★ 15. Floor and ceiling are to calculate while analyzing algorithms

- ★ a. Very easy
- ★ b. Usually considered difficult P-31
- ★ c. 3rd Option is missing
- ★ d. 4th Option is missing

★ 16. _ of reference is an important fact of current processor technology.

- ★ a. Defining
- ★ b. Assigning
- ★ c. Formality
- ★ d. Locality P-8

★ 17. In max-heap, largest element is stored at root node. Where is the smallest element stored?

- ★ a. Right Node
- ★ b. Leaf Node
- ★ c. Middle Node
- ★ d. Left Node

★ 18. In average-case time analysis of Quick sort algorithm, the most balanced case for partition is when we divide the list of elements into _.

- ★ a. Equal no. of pieces as of input elements
- ★ b. Single piece exactly
- ★ c. Two nearly equal pieces
- ★ d. Three nearly equal pieces

★ 19. Which of the following is calculated with Big O notation?

- ★ a. medium bounds
- ★ b. Upper bounds Page - 25
- ★ c. Lower bounds

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

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d. Both upper and lower bounds

20. Edit distance algorithm based on strategy

a. Greedy

b. Dynamic Programming Page - 81

c. Divide and Conquer

d. Searching

21. In Heapsort Algorithm, total time taken by heapify procedure is _

a. $O(\log n)$ Page-43

b. $O(\log^2 n)$

c. $O(n \log n)$

d. $O(n^2 \log n)$

22. Al-Khwarizmi was a/an

a. Artist

b. Mathematician P-7

c. Astronomer

d. Khalifah

23. When matrix A of 5×3 is multiply with metric B of 3×4 then the number of multiplications required is: Not found exactly

a. 15

b. 12

c. 36

d. 60 Not Found exactly but as per formula at page 84,

24. Pseudo code of algorithms are to be read by _ .

a. People Page -12

b. RAM

c. Computer

d. Compiler

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25. The sieve technique is a special case, where the number of sub-problems is Just

a. 1 P-34

b. 2

c. 3

d. 4

26. When a recursive algorithm revisits the same problem over and over again, we say that the optimization problem has sub-problems.

a. Overlapping – Google Search

b. Over costing

c. Optimized

d. Three

27. Sieve technique is very important special case of Divide-and-Conquer strategy.

a. True P-34

b. False

28. In order to say anything meaningful about our algorithms, it will be important for us to settle on a .

a. Java Program

b. C++ Program

c. Pseudo program

d. Mathematical model of computation P-10

29. Merge sort is based on .

a. Brute-force

b. Plan-sweep

c. Axis-sweep

d. Divide and Conquer P-27

30. What time does Merge Sort algorithm take in order to sort an array of 'n' numbers?

a. $\Theta(n)$

b. $\Theta(\log n)$

c. $\Theta(n^2)$

d. $\Theta(n \log n)$ Google Search

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31. In Heap Sort algorithm, the first step is to _ _.
- a. Call Build-Heap procedure Page - 46
 - b. Sort the array in descending order
 - c. Call Heapify procedure
 - d. Find the number of input elements
32. The definition of theta-notation relies on proving asymptotic bound.
- a. One
 - b. Lower
 - c. Upper
 - d. Both lower & upper Page - 25
33. In merge sort algorithm, to merge two lists of size $n/2$ to a list of size n , takes _ time.
- a. Theta (n) Page - 32
 - b. Theta $\log(n)$
 - c. Theta $\log^2(n)$
 - d. Theta $n \log(n)$
34. We can make _ recursive calls in Fibonacci Sequence.
- a. Infinite
 - b. Finite google
 - c. Only one
 - d. Zero
35. Following is NOT the application of Edit Distance problem.
- a. Speech recognition
 - b. Spelling Correction
 - c. Ascending Sort Page - 76
 - d. Computational Molecular Biology
36. In plane sweep approach, a vertical line is swept across the 2d-plane and structure is used for holding the maximal points lying to the left of the sweep line.
- a. Array
 - b. Queue
 - c. Stack Page - 18
 - d. Tree
37. When a heapify procedure is applied to the root node to restore the heap, then at each level, the comparison performed takes time:

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- ★ a. It will take $(\log n)$.
- ★ b. It cannot be predicted
- ★ **c. It will take $O(1)$. Page - 43**
- ★ d. Time will vary according to the nature of input data.
- ★ 38. _ time is the maximum running time over all legal inputs.
- ★ **a. Worst-case Page - 13**
- ★ b. Average-case
- ★ c. Best-case
- ★ d. Good-case
- ★ 39. Efficient algorithm requires less computational...
- ★ a. Memory
- ★ b. Running Time
- ★ **c. Memory and Running Time Page - 9**
- ★ d. Energy
- ★ 40. For average-case time analysis of Quick sort algorithm, Pivot selection is on average basis from ____
- ★ a. half of the input values
- ★ **b. all possible random values Page – 50**
- ★ c. Pivot is input separately
- ★ d. values greater than 5
- ★ 41. Selection algorithm takes theta
- ★ a. (n^2)
- ★ **b. (n) Page - 37**
- ★ c. $\log(n)$
- ★ d. $n \log(n)$
- ★ 42. Recurrence can be described in terms of a tree.
- ★ **a. Yes Page - 31**
- ★ b. No
- ★ 43. Time complexity of Dynamic Programming based algorithm for computing the minimum cost of Chain Matrix Multiplication is
- ★ a. $\log n$
- ★ b. n
- ★ c. n^2 (n square)

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d. n^3 (n cube) Page -90

44. The Iteration method is used for

a. Comparing sorting algorithms only

b. Solving Recurrence relations Page 31

c. Merging elements in Merge sort

d. Dividing elements in Merge sort

45. In 3-Dimensional space, a point P has coordinate(s).

a. (X, Y)

b. (X, 0)

c. (0, Y)

d. (X, Y, Z)

46. Chain matrix multiplication problem can be solved through strategy.

a. Dynamic programming Page – 85

b. Greedy

c. Divide and conquer

d. Sorting

47. Merge sort have running time running time of Heap sort. Not found exactly

a. Greater than

b. Less than Google

c. Equal to

d. Different than

48. Median is not useful measure of central tendency of given input set especially when the distribution of values is highly skewed.

a. True

b. False Page – 34

49. We do not need to mathematically prove that for comparison-based sorting algorithms always takes Omega $n \log(n)$ time.

a. True Google & VU Tech (pg 46 not very clear)

b. False

50. The Omega-notation allows us to state only the asymptotic bounds.

a. Middle

b. Lower Page 25

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- ★ c. Upper
- ★ d. Both lower & upper
- ★ 51. Both lower & upper Sorting can be in
 - ★ a. Increasing order only
 - ★ b. Decreasing order only
 - ★ **c. Both Increasing and Decreasing order – (My opinion)**
 - ★ d. Random order
- ★ 52. Radix sort performs sorting the numbers digit (s) at a time.
 - ★ **a. One Page - 71**
 - ★ b. Two
 - ★ c. Three
 - ★ d. All
- ★ 53. Quicksort is a/an and sorting algorithm.
 - ★ a. Not in place, not stable one
 - ★ **b. In place , not stable one Page - 54**
 - ★ c. In place , stable one
 - ★ d. Not in place , stable one
- ★ 54. Consider three matrices X,Y,Z of dimensions 1×2 , 2×3 , 3×4 respectively. The number of multiplications of (XY) Z is:
 - ★ **a. 18 As per lecture slides**
 - ★ b. 32
 - ★ c. 24
 - ★ d. 30
- ★ 55. In Fibonacci Sequence, unnecessary repetitions do not exist at all.
 - ★ a. True
 - ★ **b. False Page – 74**
- ★ 56. It is not a Fibonacci sequence . 1,1,1,2,3,5,8,13,21,34,55,.....
 - ★ **a. True Page - 73**
 - ★ b. False
- ★ 57. Heap sort is a/ an and _ sorting algorithm.
 - ★ a. Not in place, not stable one

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b. In place , not stable one Page - 54

c. In place , stable one

d. Not in place , stable one

58. Identify the True Statement

a. The knapsack problem does not belong to the domain of optimization problems.

b. The knapsack problem belongs to the domain of optimization problems. Page - 91

c. The Knapsack problem cannot be solved by using dynamic programming

d. The knapsack problem is optimally solved by using brute force algorithm.

59. In Dynamic Programming, our approach is to _

a. Develop the solution in a top-down fashion

b. Express the problem non-recursively

c. Build the solution in a bottom-up fashion Page - 75

d. Input several sub-problems simultaneously

60. Counting sort is suitable to sort the elements in range 1 to K;

a. K is large

b. K is small Page - 57

c. K may be large or small

d. None

61. We can multiply two matrices A and B only when they are compatible which means

a. Number of columns in A must be equal to number of rows in B. it seems Correct as per page 84

b. Number of rows and columns do not matter

c. Number of columns in A must be equal to number of columns in B

d. Number of rows in A must be equal to number of rows in B

62. Matrix multiplication is a (n) operation.

a. Commutative

b. Associative Page 85

c. Neither commutative nor associative

d. Commutative but not associative

63. In Dynamic Programming approach, solution is modified / changed

a. Always once

b. At each stage google and wikipedia

c. Only for specific problems

d. At 4th stage only

64. In Knapsack problem, the goal is to put items in the Knapsack such that the value of the items is _ subject to weight limit of knapsack.

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- ★ a. Minimized
- ★ b. Decreased
- ★ **c. Maximized Page - 91**
- ★ d. None of the given options
- ★ 65. An in-place sorting algorithm is one that uses additional array for storage.
- ★ a. Always
- ★ b. Permanently
- ★ **c. Does not Page - 54**
- ★ d. Sometime
- ★ 66. Memoization is a part of Dynamic Programming Strategy.
- ★ **a. True Page - 74**
- ★ b. False
- ★ 67. If matrix A of dimension 2×4 is multiply with matrix B of dimension 4×3 , then the dimension of resultant matrix is Not found exactly
- ★ a. 2×4
- ★ b. 4×3
- ★ c. 3×4
- ★ **d. 2×3 It seems correct as per second last Para of page 84**

Quiz no 1

- ★ Which method is preferable for dealing with chain matrix multiplication?
- ★ i. Divide and conquer strategy
- ★ **ii. Dynamic programming formulation**
- ★ iii. Graph theory
- ★ iv. Greedy Approach
- ★ Huffman algorithm produces the.....prefix code tree.
- ★ i. Better
- ★ **ii. Optimal**
- ★ iii. Worst
- ★ iv. Best
- ★ A....w is adjacent to vertex v if there is an edge from v to w.
- ★ i. Acyclic
- ★ **ii. Vertex**
- ★ iii. Loop
- ★ iv. Cycle

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Using ASCII standard the string “greedy” will be encoded with

- i. 44 bits
- ii. 120 bits
- iii. 40 bits
- iv. 48 bits

In activity scheduling algorithm, each activity is represented by a

- i. Rectangle
- ii. Square
- iii. Circle
- iv. Triangle

Those problems in which greedy finds good, but not always best is called a greedy.....

- i. Heuristic
- ii. Solution
- iii. Result
- iv. Algorithm

The knapsack problem belongs to the domain of.....Problems

- i. Searching
- ii. Sorting
- iii. Linear solution
- iv. Optimization

The general coin change problem can be solved using

- i. Recursion
- ii. Greedy algorithm
- iii. Dynamic programming
- iv. Divide and conquer

Huffman algorithm produces the.....prefix code tree

- i. Optimal
- ii. Better
- iii. Best
- iv. Worst

Huffman algorithm generates an optimumcode

- i. Postfix
- ii. Infix
- iii. None of the given options
- iv. Prefix

There areways of representing graphics

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i. 2

ii. 1

iii. 3

iv. 4

Knapsack word originates fromlanguage

i. German

ii. English

iii. French

iv. Norwegian

Graphs are important.....model for many application problems

i. Mathematical

ii. Unpredictable

iii. Haphazard

iv. Unsystematic

Which type of algorithm is harder to prove the correctness?

i. Dynamic

ii. Greedy

iii. Divide and conquer

iv. Brute force

..... Items are not allowed in 0/1 knapsack problem

i. Fractional

ii. 0

iii. 1

iv. 0/1

Matrix multiplication is a(n) operation

i. Neither commutative nor associative

ii. Transitive

iii. Commutative

iv. Associative

For a Diagraph $G = (V, E)$, Sum of in-degree (v) -----.

• Not equal to sum of out-degree(v)

• = sum of out-degree(v) pg#115

• < sum of out-degree(v)

• > sum of out-degree(v)

DFS or BFS yields a-----of the graph.

• Traversed tree

• Spanning tree pg#125

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- Simple tree
- Free tree

Using ASCII code, each character is represented by a fixed-length code of ----- bits per character.

- 4
- 6
- **8 pg#100**
- 10

In Knapsack Problem, the goal is to put items in the Knapsack such that the value of the items is-----subject to weight limit of the Knapsack.

- Minimized
- Decreased
- **Maximized pg#109**
- None of the above given

A graph is said to be acyclic if it contains-----.

- At least one cycle
- Exactly one cycle
- Always more than one cycle
- **No cycles pg#116**

The number of edges that come out of a vertex is called the-----of that vertex in the digraph.

- Post-degree
- in-degree
- **out-degree pg#114**
- pre-degree

If Matrix-A has dimensions “3x2” and Matrix-B has dimensions “2x3”, then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

- 3x2
- 2x3
- 2x2
- **3x3pg#87**

A/an-----is one in which you want to find, not just a solution, but the best solution.

- **Optimization problem**
- Divide and Conquer
- NP complete problem
- Best problem

Fractional Knapsack is founded on-----method.

- **Greedy**

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- Recursive
- Divide and Conquer
- Dynamic programming

If the graph is represented using an adjacency matrix, then Breadth-first search takes-----time.

- $O(E+1)$
- **$O(V^2)$**
- $O(V)$
- $O(E)$

In inductive approach of Knapsack problem, we consider 2 cases, -----or-----.

- Median, Mode
- Recursive, Iterative
- **Leave object, Take object pg#93**
- Sequentially, Parallel

A Greedy algorithm can NOT be used to solve all the-----problems.

- **Dynamic programming (Google)**
- Memorization programming
- Edit-distance programming
- Storing value programming

In Huffman encoding, the-----is the number of occurrences of a character divided by the total characters in the message.

- Counting
- Parsing
- **Relative Probability pg#100**
- Weight

The Binary Tree constructed by a Huffman Encoding is a:

- **Full Binary Tree pg#102**
- Partial Binary Tree
- Incomplete Binary Tree
- None of the given option

Following is not the application of Edit Distance Problem.

- **Speech recognition pg#76**
- Spelling correction
- Ascending order
- Computational Molecular Biology

Consider three Matrices X, Y, Z of dimensions 1×2 , 2×3 , 3×4 respectively. The number of multiplication of (XYZ) is:

- **18**

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- 32
- 24
- 30

18) In ----- Knapsack Problem, limitation is that an item can either be put in the bag or not. Fractional items are not allowed.

- 0
- 1

• **0/1 pg#91**

- Fractional

An in-place sorting algorithm is one that-----uses additional array for storage.

- Always
- Permanently

• **Does not pg#54**

- Sometime

If Matrix-A has dimensions “ pxq ” and Matrix-B has dimensions “ qxr ”, then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

- $P \times q$

• **$P \times r$ page#84**

- $q \times r$
- $q \times p$

Counting sort is suitable to sort the elements in range 1 to K.

- K is large

• **K is small pg#71**

- K may be large or small
- None

When matrix A of 5×3 is multiply with matrix B of 3×4 then the multiplication required is:

- 15
- 12
- 36
- **60**

-----is a linear time sorting algorithm.

- Merge sort
- Quick sort
- Bubble sort

• **Radix sort**

In Dynamic Programming approach, we do not store the solution to each sub problem in case if

- True

• **False page#75**

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Dynamic Programming approach is usually useful in solving optimization problem.

- **True page#97**

- False

Which of the following algorithm provides an optimal solution for the activity selection problem?

- Divide and Conquer

- Brute force

- **Greedy pg#105**

- Recursive

A graph is----- if every vertex can reach every other vertex.

- **Connected pg#116**

- Cycle

- Acyclic

- Loop

In a Huffman encoding when a new node is created by combining two nodes, the new node is placed in the _____.

- **Priority queue pg#100**

- Linked list

- Min heap tree

- Graph traversal

Huffman algorithm produces the _____ prefix code

- **Optimal pg#105**

- Best

- Worst

- Better

In _____ algorithm, you hope that by choosing a local optimum at each step, you will end up a global optimum.

- Simple

- Divide and conquer

- **Greedy pg#97**

- Brute Force

The string "Imnede" is coded with ASCII code, the message length would be _____ bits.

- 24

- 36

- **48**

- 60

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For graph traversal, breadth-first search strategy _____

- Is always recursive
- Cannot be recursive
- Cannot be non-recursive

• **Can be both recursive and non-recursive pg#119**

In activity scheduling algorithm, the width of a rectangle _____

- Is always ignored
- Directs towards recursion
- Should be maximized

• **Indicates the duration of an activity pg#106**

If the graph is represented using an adjacency list, then Breadth-first search takes----- time

- $O(V^2)$
- $O(V)$

• **$O(V+E)$ pg#138**

- $O(E+1)$

Suppose you are given infinite coins of 1,2 ,3, and 4. Select the ways of the minimum number of coins that required to achieve a sum of 6:

- 1
- **2 conceptual**
- 3
- 4

Using ASCII standard the string “greedy” will be encoded with

• **48 bits Conceptual**

- 120 bits
- 44 bits
- 40 bits

The Huffman codes provide a method of -----data efficiency.

- Reading/Writing
- **Encoding/Decoding pg#99**
- Divide/Conquer
- Inserting/Deleting

In the context of activity selection algorithm, time is dominated by sorting of the activities by-----

- Start Times
- **Finish Times pg#106**
- Average Times
- CPU Burst Times

Time complexity of the “0-1” knapsack algorithm depends on----

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- Number of items
- Capacity of the knapsack
- Size of the Table

• Number of items and capacity of knapsack (not sure)

The greedy approach gives us an optimal solution when the coins are all powers of a ----- denomination

• Fixed pg#98

- Variable
- Constant
- Static

In Activity Selection, we say that two activities are non-interfering if their start-finish interval-----overlap

• Do not pg#105

- Sometimes
- Once

How many steps are involved to design the dynamic programming strategy?

- 2
 - 3
 - 1
- ## • 4 pg#92

Bag is a-----

• type of algorithm pg#119

- data structure
- program
- compiler

The running time of brute-force algorithm to solve Knapsack problem is-----

- $O(n)$
 - $O(n^3)$
 - $O(n!)$
- ## • $O(2^n)$ pg#92

If a matrix has three rows and two columns, the dimension of matrix will be:

- ## • 3x2 Conceptual
- 2x3
 - 3x3
 - 2x2

The Probability in Huffman encoding is the number of occurrences of a character divided by the total-

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-----in the message.

- Numbers
- Frequencies
- Strings
- **Characters pg#100**

In recursive formulation of Knapsack Problem: $V[0, j] = \text{-----}$ for $j \geq 0$

- -1
- **0 pg#93**
- 1
- 2

The Knapsack Problem belongs to the domain of-----problem.

- **Optimization pg#91**
- Sorting
- Linear solution
- Searching

An optimization problem is one in which you want to find the-----solution.

- Simple
- Good
- **Best pg#97**
- Worst

Huffman algorithm generates an optimum----- code.

- **Prefix page 102**
- Postfix
- Infix
- None of the above

Which of the following algorithms solves the Fractional Knapsack Problem more effectively?

- Divide and Conquer
- **Greedy algorithm page 109**
- Dynamic programming
- Backtracking

If we implement the bag by using a queue, we have-----.

- **BFS page 124**
- DFS
- Graph
- Loop

One of the limitation in 0/1 Knapsack is that an item can either be-----in the bag or not.

- Use
- **Put page 91**

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- Move
- Store

In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object 'i' then we gain-----.

- W(Total weight of Knapsack)
- V(Total value of all items)
- **vi(value of object i) page 93**
- None of the given option

An activity scheduling algorithm, the width of a rectangle-----.

- Is always ignored
- Direct toward recursion
- Should be maximized
- **Indicates the duration of an activity page 106**

The prefix code generated by Huffman algorithm----- the expected length of the encoded string.

- **Minimizes page 102**
- Balances
- Maximizes
- Keeps constant

In Huffman encoding, the characters with smallest probabilities are placed at the-----depth of the tree.

- Minimum
- Average
- **Maximum page 102**
- Root

There are ----- ways to representing graphs.

- 3
- 1
- **2 page 116**
- 4

Each time we traverse graph by Breadth-first search algorithm, we count the distance from-----.

- Starting node
- **Neighbors of the starting node page 117**
- Right most node
- Left most node

In Activity Selection problem, intuitively-----.

- There are always short activities as input

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- Short activities are not attractive
- Duration of the activities does not matter
- **We do not like long activities page 105**

62) The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an-----solution.

- Simple
- Sub-optimal
- **Optimal page 105**
- Non optimal

The ----- is a problem for which the greedy algorithm approach provides an optimal solution.

• **Activity selection page 105**

- Dynamic programming
- Knapsack problem
- NP complete problem

Which of the following ways can be used to represent a graph?

- Adjacency list
- Incidence matrix
- **Adjacency list , Adjacency Matrix page 116**
- No way to represent

Queue Data Structure work on ----- principles.

- **FIFO(first in first out) cs301**
- LIFO(last in first out)
- JLO(just in last out)
- LOFI(last out first in)

Graphs can be represented by an----- and -----.

- queue , stack
- **adjacency list , adjacency matrix page 116**
- adjacency right , adjacency left
- Binary, linear

Identify a TRUE statement about Knapsack.

- The Knapsack Problem does not belongs to the domain of optimization problems
- **The Knapsack Problem belongs to the domain of optimization problems page**

91

- The Knapsack Problem cannot be solved by using Dynamic programming
- The Knapsack Problem is optimally solved by using Brute force algorithm

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Which of the following is true about graph?

- A graph may contain no edges and many vertices
- **A graph may contain many edges and no vertices**

Which type of algorithm is harder to prove the correctness?

- **Greedy google**
- Brute force
- Dynamic programming
- Divide and Conquer

Final term Quiz

1. Which method is preferable for dealing with chain matrix multiplication?

- ☐ Divide and conquer strategy
- ☒ **Dynamic programming formulation**
- ☐ Graph theory
- ☐ Greedy Approach

2. Huffman algorithm produces the.....prefix code tree.

- ☐ Better
- ☒ **Optimal**
- ☐ Worst
- ☐ Best

3. A....w is adjacent to vertex v if there is an edge from v to w.

- ☐ Acyclic
- ☒ **Vertex**
- ☐ Loop
- ☐ Cycle

Using ASCII standard the string "greedy" will be encoded with

- i. 44 bits
- ii. 120 bits
- iii. 40 bits
- iv. 48 bits**

Find the maximum value of the items which can carry using knapsack weight capacity =50

Item weight Value

- 10 70
- 20 20
- 30 80
- 70 200
- i. 90
- ii. 280

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iii. 200

iv. 100

In activity scheduling algorithm, each activity is represented by a

Rectangle

ii. Square

iii. Circle

iv. Triangle

Those problems in which greedy finds good, but not always best is called a greedy.....

i. Heuristic

ii. Solution

iii. Result

iv. Algorithm

The knapsack problem belongs to the domain of.....Problems

i. Searching

ii. Sorting

iii. Linear solution

iv. Optimization

The general coin change problem can be solved using

i. Recursion

ii. Greedy algorithm

iii. Dynamic programming

iv. Divide and conquer

Huffman algorithm produces the.....prefix code tree

i. Optimal

ii. Better

iii. Best

iv. Worst

Huffman algorithm generates an optimumcode

i. Postfix

ii. Infix

iii. None of the given options

iv. Prefix

There areways of representing graphics

i. 2

ii. 1

iii. 3

iv. 4

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Knapsack word originates fromlanguage

i. German

ii. English

iii. French

iv. Norwegian

Graphs are important.....model for many application problems

i. Mathematical

ii. Unpredictable

iii. Haphazard

iv. Unsystematic

Which type of algorithm is harder to prove the correctness?

i. Dynamic

ii. Greedy

iii. Divide and conquer

iv. Brute force

.....Items are not allowed in 0/1 knapsack problem

i. Fractional

ii. 0

iii. 1

iv. 0/1

Matrix multiplication is a(n).....operation

i. Neither commutative nor associative

ii. Transitive

iii. Commutative

iv. Associative

For a Diagraph $G = (V, E)$, Sum of in-degree (v) -----.

• Not equal to sum of out-degree(v)

• **= sum of out-degree(v) pg#115**

• < sum of out-degree(v)

• > sum of out-degree(v)

DFS or BFS yields a-----of the graph.

• Traversed tree

• **Spanning tree pg#125**

• Simple tree

• Free tree

Using ASCII code, each character is represented by a fixed-length code of-----bits per character.

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- 4
- 6
- **8 pg#100**
- 10

In Knapsack Problem, the goal is to put items in the Knapsack such that the value of the items is-----
-----subject to weight limit of the Knapsack.

- Minimized
- Decreased
- **Maximized pg#109**
- None of the above given

Consider the string "abcedaacac", if the string is coded with ASCII codes, the message length would be---

- 70 bits
- 60 bits
- 90 bits
- **72 bits**

A graph is said to be acyclic if it contains-----.

- At least one cycle
- Exactly one cycle
- Always more than one cycle
- **No cycles pg#116**

The number of edges that come out of a vertex is called the-----of that vertex in the digraph.

- Post-degree
- in-degree
- **out-degree pg#114**
- pre-degree

If Matrix-A has dimensions "3x2" and Matrix-B has dimensions "2x3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

- 3x2
- 2x3
- 2x2
- **3x3pg#87**

A/an-----is one in which you want to find, not just a solution, but the best solution.

- **Optimization problem**
- Divide and Conquer
- NP complete problem
- Best problem

Fractional Knapsack is founded on-----method.

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- **Greedy**

- Recursive
- Divide and Conquer
- Dynamic programming

If the graph is represented using an adjacency matrix, then Breadth-first search takes-----time.

- $O(E+1)$
- **$O(V^2)$**

- $O(V)$
- $O(E)$

In inductive approach of Knapsack problem, we consider 2 cases, -----or-----.

- Median, Mode
- Recursive, Iterative
- **Leave object, Take object pg#93**
- Sequentially, Parallel

A Greedy algorithm can NOT be used to solve all the-----problems.

- **Dynamic programming (Google)**
- Memorization programming
- Edit-distance programming
- Storing value programming

In Huffman encoding, the-----is the number of occurrences of a character divided by the total characters in the message.

- Counting
- Parsing
- **Relative Probability pg#100**
- Weight

The Binary Tree constructed by a Huffman Encoding is a:

- **Full Binary Tree pg#102**
- Partial Binary Tree
- Incomplete Binary Tree
- None of the given option

Following is not the application of Edit Distance Problem.

- Speech recognition pg#76
- Spelling correction
- Ascending order
- Computational Molecular Biology

Consider three Matrices X, Y, Z of dimensions 1×2 , 2×3 , 3×4 respectively. The number of multiplication of (XYZ) is:

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• 18

• 32

• 24

• 30

18) In ----- Knapsack Problem, limitation is that an item can either be put in the bag or not.

Fractional items are not allowed.

• 0

• 1

• 0/1 pg#91

• Fractional

An in-place sorting algorithm is one that-----uses additional array for storage.

• Always

• Permanently

• Does not pg#54

• Sometime

If Matrix-A has dimensions "pxq" and Matrix-B has dimensions "qxr", then multiplication of Matrix-

A and Matrix-B will result a new Matrix-C having dimensions.

• P x q

• P x r page#84

• q x r

• q x p

Counting sort is suitable to sort the elements in range 1 to K.

• K is large

• K is small pg#71

• K may be large or small

• None

When matrix A of 5x 3 is multiply with matrix B of 3x 4 then the multiplication required is:

• 15

• 12

• 36

• 60

-----is a linear time sorting algorithm.

• Merge sort

• Quick sort

• Bubble sort

• Radix sort

In Dynamic Programming approach, we do not store the solution to each sub problem in case if it

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reappears.

- True
- False pg#75

Dynamic Programming approach is usually useful in solving optimization problem.

- True page#97
- False

Which of the following algorithm provides an optimal solution for the activity selection problem?

- Divide and Conquer
- Brute force
- Greedy pg#105

A graph is----- if every vertex can reach every other vertex.

- Connected pg#116
- Cycle
- Acyclic
- Loop

In a Huffman encoding when a new node is created by combining two nodes, the new node is placed in the _____.

- Priority queue pg#100
- Linked list
- Min heap tree
- Graph traversal

Huffman algorithm produces the _____ prefix code

- Optimal pg#105
- Best
- Worst
- Better

In _____ algorithm, you hope that by choosing a local optimum at each step, you will end up a global optimum.

- Simple
- Divide and conquer
- Greedy pg#97
- Brute Force

The string "Imncde" is coded with ASCII code, the message length would be ____ bits.

- 24

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- 36
- **48**
- 60

For graph traversal, breadth-first search strategy _____

- Is always recursive
- Cannot be recursive
- Cannot be non-recursive
- Can be both recursive and non-recursive page 119

In activity scheduling algorithm, the width of a rectangle _____

- Is always ignored
- Directs towards recursion
- Should be maximized
- Indicates the duration of an activity pg#106

If the graph is represented using an adjacency list, then Breadth-first search takes----- time

- $O(V^2)$
- $O(V)$
- $O(V+E)$ pg#138
- $O(E+1)$

Suppose you are given infinite coins of 1,2,3, and 4. Select the ways of the minimum number of coins that required to achieve a sum of 6:

- 1
- 2 conceptual

Using ASCII standard the string "greedy" will be encoded with

- 48 bits Conceptual
- 120 bits
- 44 bits
- 40 bits

The Huffman codes provide a method of -----data efficiency.

- Reading/Writing
- Encoding/Decoding pg#99
- Divide/Conquer
- Inserting/Deleting

In the context of activity selection algorithm, time is dominated by sorting of the activities by-----

- Start Times
- Finish Times pg#106
- Average Times
- CPU Burst Times

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Time complexity of the “0-1” knapsack algorithm depends on----

- Number of items
- Capacity of the knapsack
- Size of the Table
- Number of items and capacity of knapsack (not sure)

The greedy approach gives us an optimal solution when the coins are all powers of a -----
Denomination

- Fixed pg#98
- Variable
- Constant
- Static

In Activity Selection, we say that two activities are non-interfering if their start-finish interval-----
overlap

- Do
- Do not pg#105
- Sometimes
- Once

How many steps are involved to design the dynamic programming strategy?

- 2
- 3
- 1
- 4 pg#92

Bag is a-----

- type of algorithm pg#119
- data structure
- program
- compiler

1. If a problem is in NP-complete, it must also be in NP.

- True page#170
- False

2. The Huffman algorithm finds a optimal solution.

- True pg#105
- False

3. The Huffman algorithm finds an exponential solution

- True

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► False pg#105

4. The Huffman algorithm finds a polynomial solution

► True google

► False

5. The greedy part of the Huffman encoding algorithm is to first find two nodes with smallest frequency.

► True pg#100

► False

6. The code word assigned to characters by the Huffman algorithm have the property that no code word is the prefix of any other.

► True pg#101

► False

7. Huffman algorithm uses a greedy approach to generate a postfix code T that minimizes the expected length B (T) of the encoded string.

► True

► False pg#102

8. Dijkstra's single source shortest path algorithm works if all edges weights are non-negative and there are negative cost cycles.

► True

► False pg#154

9. The term "coloring" came from the original application which was in architectural design.

► True

► False pg#176

10. In the clique cover problem, for two vertices to be in the same group, they must be adjacent to each other.

► True pg#176

► False

11. Dijkstra's algorithm is operated by maintaining a subset of vertices

► True pg#155

► False

12. We do sort to,

► keep elements in random positions

► keep the algorithm run in linear order

► keep the algorithm run in $(\log n)$ order

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► keep elements in increasing or decreasing order pg#40

13. After partitioning array in Quick sort, pivot is placed in a position such that

► Values smaller than pivot are on left and larger than pivot are on right pg#48

► Values larger than pivot are on left and smaller than pivot are on right

☐ Pivot is the first element of array

► Pivot is the last element of array

14. Merge sort is stable sort, but not an in-place algorithm

► True pg#54

► False

15. A $p \times q$ matrix A can be multiplied with a $q \times r$ matrix B. The result will be a $p \times r$ matrix C. There are $(p \cdot r)$ total entries in C and each takes _____ to compute.

► $O(q)$ pg#84

► $O(1)$

► $O(n^2)$

► $O(n^3)$

16. One of the clever aspects of heaps is that they can be stored in arrays without using any

☐ Pointers pg#40

☐ constants

☐ variables

☐ functions

17. Merge sort requires extra array storage,

☐ True pg#54

☐ False

18. The Huffman codes provide a method of encoding data inefficiently when coded using ASCII standard.

☐ True

☐ False pg#99

19. Using ASCII standard the string abacdaacac will be encoded with _____ bits.

☐ 80 pg#99

☐ 160

☐ 320

☐ 100

20. Using ASCII standard the string abacdaacac will be encoded with 160 bits.

☐ True

☐ False pg#99

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(03066793776)

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21. Using ASCII standard the string abacdaacac will be encoded with 10 bytes.

- ☐ True
- ☐ False pg#99

23. Huffman algorithm uses a greedy approach to generate an prefix code T that minimizes the expected length B (T) of the encoded string.

- ☐ True pg#100
- ☐ False

24. An optimization problem is one in which you want to find,

- ▶ Not a solution
- ▶ An algorithm
- ▶ Good solution
- ▶ The best solution pg#97

25. Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a large number of vert

- ▶ True pg#149
- ▶ False

26. If a problem is in NP, it must also be in P.

- ▶ True
- ▶ False
- ▶ unknown pg#173

27. What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

- ▶ Lists require less space than matrices but take longer to find the weight of an edge (v1,v2) pg#116
- ▶ Lists require less space than matrices and they are faster to find the weight of an edge (v1,v2)
- ▶ Lists require more space than matrices and they take longer to find the weight of an edge (v1,v2)

If a graph has v vertices and e edges then to obtain a spanning tree we have to delete

- ▶ v edges.
- ▶ $v - e + 5$ edges
- ▶ $v + e$ edges.
- ▶ **None of these**

28. Maximum number of vertices in a Directed Graph may be $|V|^2$

- ▶ True
- ▶ False pg#115

29. The Huffman algorithm finds a (n) _____ solution.

- ▶ Optimal pg#105
- ▶ non-optimal

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(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

- ▶ Exponential
- ▶ Polynomial

30. The Huffman algorithm finds an exponential solution

- ▶ True
- ▶ False pg#115

31. Edge (u, v) is a forward edge if

- ▶ u is a proper descendant of v in the tree
- ▶ v is a proper descendant of u in the tree pg#129
- ▶ None of these

32. After partitioning array in Quick sort, pivot is placed in a position such that

- ▶ Values smaller than pivot are on left and larger than pivot are on right (Page 48)
- ▶ Values larger than pivot are on left and smaller than pivot are on right
- ▶ Pivot is the first element of array
- ▶ Pivot is the last element of array

33. Merge sort is stable sort, but not an in-place algorithm

- ▶ True Page #54
- ▶ False

34. In counting sort, once we know the ranks, we simply _____ numbers to their final positions in an output array.

- ▶ Delete
- ▶ copy Page# 57
- ▶ Mark
- ▶ arrange

35. Dynamic programming algorithms need to store the results of intermediate sub-problems.

- ▶ True pg#75
- ▶ False

36. A $p \times q$ matrix A can be multiplied with a $q \times r$ matrix B. The result will be a $p \times r$ matrix C. There are (p . r) total entries in C and each takes _____ to compute.

- ▶ $O(q)$ pg#48
- ▶ $O(1)$
- ▶ $O(n^2)$
- ▶ $O(n^3)$

37. _____ is a graphical representation of an algorithm

- ▶ notation
- ▶ notation
- ▶ **Flowchart**

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(03066793776)

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► Asymptotic notation

38. Which of the following is calculated with big o notation?

- ☐ ► Lower bounds
- ☐ ► Upper bounds pg#25
- ☐ ► Both upper and lower bound
- ☐ ► Medium bounds

39. Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?

- ☐ ► The array elements form a heap
- ☒ ► **Elements in each half of the array are sorted amongst themselves**
- ☐ ► Elements in the first half of the array are less than or equal to elements in the second half of the array
- ☐ ► None of the above

40. What is the solution to the recurrence $T(n) = T(n/2) + n$, $T(1) = 1$

- ☐ ► $O(\log n)$
- ☐ ► $O(n)$ pg#37

41. Consider the following Huffman Tree

The binary code for the string TEA is

- ☒ ► **10 00 010**
- ☐ ► 011 00 010
- ☐ ► 10 00 110
- ☐ ► 11 10 110

42. A greedy algorithm does not work in phases.

- True
- False pg#97

43. Can an adjacency matrix for a directed graph ever not be square in shape?

- ☐ ► Yes
- ☐ ► No pg#116

44. One of the clever aspects of heaps is that they can be stored in arrays without using any_____.

- ☐ ► Pointers pg#40
- ☐ ► constants
- ☐ ► variables
- ☐ ► functions

45. Merge sort requires extra array storage,

- ☐ ► True Page 54
- ☐ ► False

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46. Non-optimal or greedy algorithm for money change takes _____

- ☐ ► $O(k)$ Pg #99
- ☐ ► $O(kN)$
- ☐ ► $O(2k)$
- ☐ ► $O(N)$

47. The Huffman codes provide a method of encoding data inefficiently when coded using ASCII standard.

- ☐ ► True
- ☐ ► False Pg# 99

48. Using ASCII standard the string abacdaacac will be encoded with _____ bits.

- ☐ ► 80 pg# 99
- ☐ ► 160
- ☐ ► 320
- ☐ ► 100

49. Using ASCII standard the string abacdaacac will be encoded with 160 bits.

- ☐ ► True
- ☐ ► False (Pg# 99)

50. Using ASCII standard the string abacdaacac will be encoded with 320 bits.

- ☐ ► True
- ☐ ► False (Pg# 99)

51. Using ASCII standard the string abacdaacac will be encoded with 100 bits.

- ☐ ► True
- ☐ ► False (Pg# 99)

52. Using ASCII standard the string abacdaacac will be encoded with 32 bytes

- ☐ ► True
- ☐ ► False (Pg# 99) 8bytes

53. The greedy part of the Huffman encoding algorithm is to first find two nodes with smallest frequency.

- ☐ ► True (Pg# 100)
- ☐ ► False

54. The greedy part of the Huffman encoding algorithm is to first find two nodes with character frequency

- ☐ ► True
- ☐ ► False (Pg# 100)

55. Huffman algorithm uses a greedy approach to generate an antefix code T that minimizes the expected length $B(T)$ of the encoded string.

- ☐ ► True

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FOR QUIZ CONTECT
(03066793776)

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☐ ► False (Pg# 102)

56. Depth first search is shortest path algorithm that works on un-weighted graphs.

► True

► False (Pg# 159)

57. Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of the algorithm is in the clever recursive formulation of the shortest path problem.

► True (Pg# 162)

► False

58. Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursive evaluation by generating a table for

☐ ► k

☐ ► k

ij d (Pg# 164)

☐ ► True

☐ ► False

59. The term coloring came from the original application which was in map drawing.

☐ ► True (Pg# 176)

☐ ► False

60. In the clique cover problem, for two vertices to be in the same group, they must be _____ each other.

☐ ► Apart from

☐ ► Far from

☐ ► Near to

☐ ► Adjacent to (Pg# 176)

61. Fixed-length codes may not be efficient from the perspective of _____ the total quantity of data.

Select correct option:

► Minimizing Pg# 99

► Averaging

► Maximizing

► Summing

62. In greedy algorithm, at each phase, you take the _____ you can get right now, without regard for future consequences.

► Worst

► Minimum

► Good

► Best Page #97

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FOR QUIZ CONTACT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

63. The difference between Prim's algorithm and Dijkstra's algorithm is that Dijkstra's algorithm uses a same key.

► True

► False Page# 156

64. If a problem is in NP-complete, it must also be in NP.

► True Page# 178

► False

65. If there are n items, there are _____ possible combinations of the items.

► 2

► n

► 2^n Page# 92

► 3^n

66. Using ASCII code, each character is represented by a fixed-length code word of _____ bits per character.

► 4

► 6

► 8 pg # 99

► 10

67. In Knapsack Problem, the thief's goal is to put items in the bag such that the _____ of the items does not exceed the limit of the bag.

► Value Page #91

► Weight

► Length

► Balance

68. The knapsack problem does not belong to the domain of optimization problems.

► True

► False Page# 91

69. In Huffman encoding, for a given message string, the frequency of occurrence (relative probability) of each character in the message is determined last.

► True

► False Page 100

70. Fixed-length codes are known for easy break up of a string into its individual characters.

► True Page# 99

► False

71. In _____ Knapsack Problem, limitation is that an item can either be put in the bag or not-fractional items

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

are not allowed.

► 0

► 1

► 0/1 Page# 91

► Fractional

72. In Knapsack Problem, value and weight both are to be under consideration.

► True page #91

► False

73. Time complexity of DP based algorithm for computing the minimum cost of chain matrix Multiplication is _____.

► $\log n$

► n

► n^2

► n^3 Page #90

74. In DP based solution of knapsack problem, to compute entries of V we will imply a/an _____ approach.

► Subjective

► Inductive Page #93

► Brute force

► Combination

75. A greedy algorithm sometimes works well for optimization problems.

► True Page# 97

► False

76. In Huffman encoding, frequency of each character can be determined by parsing the message and _____ how many times each character (or symbol) appears.

► Printing

► Incrementing

► [Counting \(Page 100\)](#)

► Deleting

77. Greedy algorithm can do very poorly for some problems.

► True Page# 97

► False

78. The Huffman codes provide a method of _____ data efficiently.

► Reading

► Encoding Page# 99

► Decoding

► Printing

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FOR QUIZ CONTECT
(03066793776)

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79. In _____ based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

- Brute force
- Dynamic programming Page #93

80. Those problems in which Greedy finds good, but not always best is called a greedy_____.

- Algorithm
- Solution
- Heuristic Page# 97
- Result

81. In brute force based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

- TRUE
- FALSE Page# 97

82. _____ problem, we want to find the best solution.

- Minimization
- Averaging
- Optimization Page# 97
- Maximization

83. Using ASCII standard the string abacdaacac will be encoded with 10 bytes.

- True Page #101
- False

84. In _____ algorithm, you hope that by choosing a local optimum at each step, you will end up at a global optimum.

- Simple
- Non-Greedy
- Greedy Page# 97
- Brute force

85. Huffman algorithm uses a greedy approach to generate an prefix code T that minimizes the expected length B (T) of the encoded string.

- True Page# 102
- False

86. Counting Money problem is an example which cannot be optimally solved by greedy algorithm.

- True Page# 97
- False

87. Huffman algorithm generates an optimum prefix code.

- True Page# 102
- False

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FOR QUIZ CONTECT
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88. If the string "lmncde" is coded with ASCII code, the message length would be _____ bits.

- ▶ 24
- ▶ 36
- ▶ $48 \times 6 = 48$ page #99
- ▶ 60

89. There are _____ nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.

- ▶ 2
- ▶ 3 Page# 90
- ▶ 4

90. A number of lectures are to be given in a single lecture hall. Optimum scheduling for this is an example of Activity selection.

- ▶ True Page# 105
- ▶ False

91. The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an _____ solution.

- ▶ Simple
- ▶ Sub optimal
- ▶ Optimal Page# 105
- ▶ Non optimal

92. The string |xyz|, if coded with ASCII code, the message length would be 24 bits.

- ▶ True ($3 \times 8 = 24$) page#99
- ▶ False

93. An application problem is one in which you want to find, not just a solution, but the _____ solution.

- ▶ Simple
- ▶ Good Page #113
- ▶ Best

94. Suppose that a graph $G = (V, E)$ is implemented using adjacency lists. What is the complexity of a breadth first traversal of G ?

- ▶ $O(|V|^2)$
- ▶ $O(|V| + |E|)$
- ▶ $O(|V|^2|E|)$
- ▶ $O(|V| + |E|)$ pg #116

95. Which is true statement?

- ▶ Breadth first search is shortest path algorithm that works on un-weighted graphs Page #153
- ▶ Depth first search is shortest path algorithm that works on un-weighted graphs.

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MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

- ▶ Both of above are true.
- ▶ None of above are true.

96. Forward edge is:

- ▶ (u, v) where u is a proper descendent of v in the tree.
- ▶ (u, v) where v is a proper descendent of u in the tree. Page #129
- ▶ (u, v) where v is a proper ancestor of u in the tree.
- ▶ (u, v) where u is a proper ancestor of v in the tree.

97. If you find yourself in maze the better traversal approach will be :

- ▶ BFS (google)
- ▶ BFS and DFS both are valid
- ▶ Level order
- ▶ DFS

98. In digraph $G=(V,E)$; G has cycle if and only if

- ▶ The DFS forest has forward edge.
- ▶ The DFS forest has back edge Page#131
- ▶ The DFS forest has both back and forward edge

99. Back edge is:

- ▶ (u, v) where v is an ancestor of u in the tree. Page# 128
- ▶ (u,v) where u is an ancestor of v in the tree.
- ▶ (u, v) where v is an predecessor of u in the tree.
- ▶ None of above

100. Cross edge is :

- ▶ (u, v) where u and v are not ancestor of one another
- ▶ (u, v) where u is ancestor of v and v is not descendent of u.
- ▶ (u, v) where u and v are not ancestor or descendent of one another Page# 129
- ▶ (u, v) where u and v are either ancestor or descendent of one another.

101. Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few edges.

- ▶ True google
- ▶ False

102. What algorithm technique is used in the implementation of Kruskal solution for the MST?

- ▶ Greedy Technique Page# 142
- ▶ Divide-and-Conquer Technique
- ▶ Dynamic Programming Technique
- ▶ The algorithm combines more than one of the above techniques

103. What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

► $O(\log E)$

► (V)

► $(V+E)$

► $O(\log V)$ Page #152

104. The relationship between number of back edges and number of cycles in DFS is,

► Both are equal

► Back edges are half of cycles

► Back edges are one quarter of cycles

► There is no relationship between no. of edges and cycles Page# 131

105. You have an adjacency list for G , what is the time complexity to compute Graph transpose G^T ?

► $(V + E)$ Page# 138

► $(V E)$

► (V)

► (V^2)

106. There is relationship between number of back edges and number of cycles in DFS

► Both are equal.

► Cycles are half of back edges.

► Cycles are one fourth of back edges.

► There is no relationship between back edges and number of cycles. Page# 131

107. A digraph is strongly connected under what condition?

► A digraph is strongly connected if for every pair of vertices $u, v \in V$, u can reach v .

► A digraph is strongly connected if for every pair of vertices $u, v \in V$, u can reach v and vice versa. Page#

135

► A digraph is strongly connected if for at least one pair of vertex $u, v \in V$, u can reach v and vice versa.

► A digraph is strongly connected if at least one third pair of vertices $u, v \in V$, u can reach v and vice versa.

108. In in-place sorting algorithm is one that uses arrays for storage :

► An additional array

► No additional array Page#54

► Both of above may be true according to algorithm

► More than 3 arrays of one dimension.

109. In stable sorting algorithm

► One array is used

► In which duplicating elements are not handled.

► More than one arrays are required.

► Duplicating elements remain in same relative position after sorting. Page# 54

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

110. Which sorting algorithm is faster :

- ▶ $O(n^2)$
- ▶ $O(n \log n)$ Page# 46
- ▶ $O(n+k)$
- ▶ $O(n^3)$

111. In Quick sort algorithm, constants hidden in $T(n \lg n)$ are

- ▶ Large
- ▶ Medium
- ▶ Not known
- ▶ Small

112. Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:

- ▶ There is explicit combine process as well to conquer the solution.
- ▶ No work is needed to combine the sub-arrays, the array is already sorted
- ▶ Merging the sub arrays
- ▶ None of above. Page# 51

113. Dijkstra's algorithm :

- ▶ Has greedy approach to find all shortest paths
- ▶ Has both greedy and Dynamic approach to find all shortest paths
- ▶ Has greedy approach to compute single source shortest paths to all other vertices Page# 154
- ▶ Has both greedy and dynamic approach to compute single source shortest paths to all other vertices.

114. Which may be stable sort:

- ▶ Bubble sort
- ▶ Insertion sort
- ▶ Both of above page# 54
- ▶ Selection sort

115. In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent _____ series in the analysis,

- ▶ linear
- ▶ arithmetic
- ▶ geometric page #37
- ▶ exponent

116. How much time merge sort takes for an array of numbers?

- ▶ $T(n^2)$
- ▶ $T(n)$ Page# 40
- ▶ $T(\log n)$
- ▶ $T(n \log n)$

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S K KHAN ACADEMY

117. Dijkstra's algorithm is operates by maintaining a subset of vertices

☐ True Page# 155

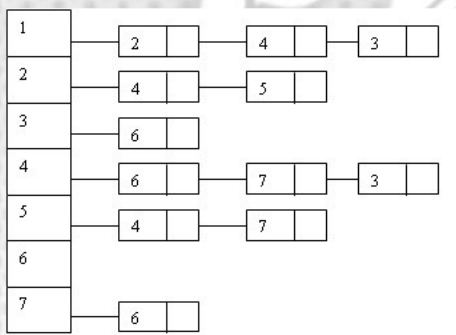
☐ False

118. The difference between Prim's algorithm and Dijkstra's algorithm is that Dijkstra's algorithm uses a different key.

► True Page# 156

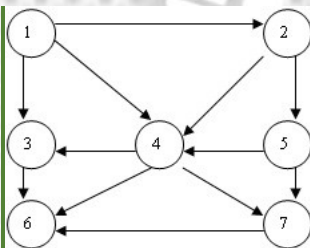
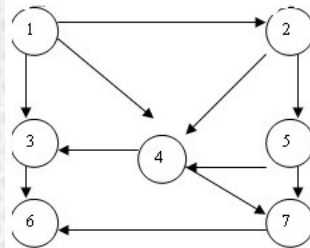
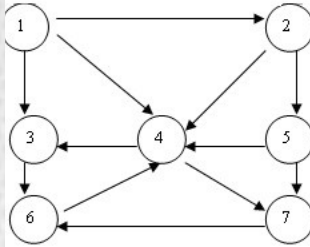
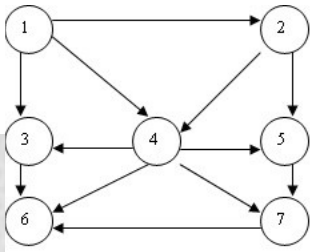
► False

118. Consider the following adjacency list:



Which of the following graph(s) describe(s) the above adjacency list?

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119. We do sorting to,

- ▶ keep elements in random positions
- ▶ keep the algorithm run in linear order
- ☐ ▶ keep the algorithm run in $(\log n)$ order
- ☐ ▶ keep elements in increasing or decreasing order Page# 40

120. After partitioning array in Quick sort, pivot is placed in a position such that

- ☐ ▶ Values smaller than pivot are on left and larger than pivot are on right Page# 48
- ☐ ▶ Values larger than pivot are on left and smaller than pivot are on right
- ☐ ▶ Pivot is the first element of array

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

☐ ► Pivot is the last element of array

121. Merge sort is stable sort, but not an in-place algorithm

☐ ► True (Page# 54)

☐ ► False

122. In counting sort, once we know the ranks, we simply _____ numbers to their final positions in an output array.

☐ ► Delete

☐ ► copy (Page# 57)

☐ ► Mark

☐ ► arrange

123. Dynamic programming algorithms need to store the results of intermediate sub-problems.

☐ ► True (Page# 75)

☐ ► False

124. A $p \times q$ matrix A can be multiplied with a $q \times r$ matrix B. The result will be a $p \times r$ matrix C. There are $(p \cdot r)$ total entries in C and each takes _____ to compute.

☐ ► $O(q)$ (Page# 84)

☐ ► $O(1)$

☐ ► $O(n^2)$

☐ ► $O(n^3)$

125. _____ is a graphical representation of an algorithm

☐ ► notation

☐ ► notation

☐ ► Flowchart

☐ ► Asymptotic notation

126. Which of the following is calculated with big o notation?

☐ ► Lower bounds

☐ ► Upper bounds (Page# 25)

☐ ► Both upper and lower bound

☐ ► Medium bounds

126. Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?

☐ ► The array elements form a heap

☒ ► **Elements in each half of the array are sorted amongst themselves**

☐ ► Elements in the first half of the array are less than or equal to elements in the second half of the array

☐ ► None of the above

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

127. non-optimal or greedy algorithm for money change takes_____

☐ ► $O(k)$ (Page #99)

☐ ► $O(kN)$

☐ ► $O(2k)$

☐ ► $O(N)$

128. The Huffman codes provide a method of encoding data inefficiently when coded using ASCII standard.

☐ ► True

☐ ► False (Page# 99)

129. Using ASCII standard the string abacdaacac will be encoded with _____ bits.

☐ ► 80 (Page #99)

☐ ► 160

☐ ► 320

☐ ► 100

130. Using ASCII standard the string abacdaacac will be encoded with 160 bits.

☐ ► True

☐ ► False (Page #99)

131. using ASCII standard the string abacdaacac will be encoded with 32 bytes

► True

► False (Page #99)

132. The greedy part of the Huffman encoding algorithm is to first find two nodes with smallest frequency.

► True (Page #100)

► False

133. The greedy part of the Huffman encoding algorithm is to first find two nodes with character frequency

► True

► False (Page# 100)

134. Huffman algorithm uses a greedy approach to generate an antefix code T that minimizes the expected length B (T) of the encoded string.

► True

► False (Page #102)

135. Depth first search is shortest path algorithm that works on un-weighted graphs.

► True

► False (Page# 153)

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

136. Dijkstras single source shortest path algorithm works if all edges weights are non negative and there are no negative cost cycles.

► True (Page#159)

► False

137. Dijkstra s single source shortest path algorithm works if all edges weights are negative and there are no negative cost cycles.

► True

► False (Page #159)

138. Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of the algorithm is in the clever recursive formulation of the shortest path problem.

► True (Page #162)

► False

139. Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursive evaluation by generating a table for

► k

► k

ij d (Page# 164)

► True

► False

140. the term coloring came from the original application which was in map drawing.

► True (Page #176)

► False

141. In the clique cover problem, for two vertices to be in the same group, they must be _____ each other.

► Apart from

► Far from

► Near to

► Adjacent to (Page# 176)

142. Fixed-length codes may not be efficient from the perspective of _____ the total quantity of data.

Select correct option:

► Minimizing (Page #99)

► Averaging

► Maximizing

► Summing

143. In greedy algorithm, at each phase, you take the _____ you can get right now, without regard for future consequences.

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

- ▶ Worst
- ▶ Minimum
- ▶ Good
- ▶ Best (Page #97)

144. The difference between Prim's algorithm and Dijkstra's algorithm is that Dijkstra's algorithm uses a same key.

- ▶ True
- ▶ False (Page# 156)

145. If a problem is in NP-complete, it must also be in NP.

- ▶ True (Page #178)
- ▶ False

146. If there are n items, there are _____ possible combinations of the items.

- ☐ ▶ 2
- ☐ ▶ n
- ☐ ▶ 2^n (Page# 92)
- ☐ ▶ 3^n

147. Using ASCII code, each character is represented by a fixed-length code word of _____ bits per character.

- ☐ ▶ 4
- ☐ ▶ 6
- ☐ ▶ 8 (Page #99)
- ☐ ▶ 10

148. In Knapsack Problem, the thief's goal is to put items in the bag such that the _____ of the items does not exceed the limit of the bag.

- ☐ ▶ Value (Page# 91)
- ☐ ▶ Weight
- ☐ ▶ Length
- ☐ ▶ Balance

149. The knapsack problem does not belong to the domain of optimization problems.

- ☐ ▶ True
- ☐ ▶ False (Page# 91)

150. In Huffman encoding, for a given message string, the frequency of occurrence (relative probability) of each character in the message is determined last.

- ☐ ▶ True
- ☐ ▶ False (Page #100)

151. Fixed-length codes are known for easy break up of a string into its individual

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

characters.

► True (Page# 99)

► False

152. In _____ Knapsack Problem, limitation is that an item can either be put in the bag or not-fractional items are not allowed.

☐ ► 0

☐ ► 1

☐ ► 0/1 (Page# 91)

☐ ► Fractional

153. The term “coloring” came from the original application which was in architectural design.

☐ ► True

☐ ► False (Page# 173)

154. In Knapsack Problem, value and weight both are to be under consideration.

☐ ► True (page# 91)

☐ ► False

155. Time complexity of DP based algorithm for computing the minimum cost of chain matrix Multiplication is _____ .

☐ ► $\log n$

☐ ► n

☐ ► n^2

☐ ► n^3 (Page# 90)

156. In DP based solution of knapsack problem, to compute entries of V we will imply a/an _____ approach.

☐ ► Subjective

☐ ► Inductive (Page# 93)

☐ ► Brute force

☐ ► Combination

157. A greedy algorithm sometimes works well for optimization problems.

☐ ► True (Page# 97)

☐ ► False

157. In Huffman encoding, frequency of each character can be determined by parsing the message and _____ how many times each character (or symbol) appears.

☐ ► Printing

☐ ► Incrementing

☐ ► Counting (Page# 100)

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

☐ ► Deleting

159. Greedy algorithm can do very poorly for some problems.

☐ ► True (Page# 97)

☐ ► False

160. The Huffman codes provide a method of _____ data efficiently.

☐ ► Reading

☐ ► Encoding (Page# 99)

☐ ► Decoding

☐ ► Printing

161. In _____ based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

☐ ► Brute force

☐ ► Dynamic programming (Page #93)

162. Those problems in which Greedy finds good, but not always best is called a greedy_____.

☐ ► Algorithm

☐ ► Solution

☐ ► Heuristic (Page# 97)

☐ ► Result

163. In brute force based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

☐ ► TRUE

☐ ► FALSE (Page# 97)

164. _____ problem, we want to find the best solution.

☐ ► Minimization

☐ ► Averaging

☐ ► Optimization (Page#97)

☐ ► Maximization

165. Using ASCII standard the string abacdaacac will be encoded with 10 bytes.

☐ ► True (Page# 101)

☐ ► False

166. In _____ algorithm, you hope that by choosing a local optimum at each step, you will end up at a global optimum.

☐ ► Simple

☐ ► Non Greedy

☐ ► Greedy (Page# 97)

☐ ► Brute force

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

167. Huffman algorithm uses a greedy approach to generate an prefix code T that minimizes the expected length B (T) of the encoded string.

☐ ▶ True (Page #102)

☐ ▶ False

168. How many elements do we eliminate in each time for the Analysis of Selection algorithm?

▶ $n / 2$ elements (Page #37)

▶ $(n / 2) + n$ elements

▶ $n / 4$ elements

▶ $2n$ elements

169. Slow sorting algorithms run in,

▶ $T(n^2)$ (Page #39)

▶ $T(n)$

▶ $T(\log n)$

▶ $T(n \log n)$

170. Counting sort is suitable to sort the elements in range 1 to k:

▶ K is large

▶ K is small (Page# 57)

▶ K may be large or small

▶ None

171. Heaps can be stored in arrays without using any pointers; this is due to the _____ nature of the binary tree,

▶ left-complete (Page# 40)

▶ right-complete

▶ tree nodes

▶ tree leaves

172. Sieve Technique can be applied to selection problem?

▶ True (Page#35)

▶ False

173. A heap is a left-complete binary tree that conforms to the _____

▶ increasing order only

▶ decreasing order only

▶ heap order (Page# 40)

▶ $(\log n)$ order

174. Divide-and-conquer as breaking the problem into a small number of

▶ pivot

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

- ▶ Sieve
- ▶ smaller sub problems (Page #34)
- ▶ Selection

175. In Sieve Technique we do not know which item is of interest

- ▶ True (Page# 34)
- ▶ False

176. The recurrence relation of Tower of Hanoi is given below $T(n) = \begin{cases} 1 & \text{if } n=1 \\ 2T(n-1) & \text{if } n > 1 \end{cases}$ In order to move a tower of 5 rings from one peg to another, how many ring moves are required?

- ▶ 16
- ▶ 10
- ▶ 32
- ▶ **31**

177. For the heap sort, access to nodes involves simple _____ operations.

- ▶ arithmetic (Page# 41)
- ▶ binary
- ▶ algebraic
- ▶ logarithmic

178. For the sieve technique we solve the problem,

- ▶ recursively (Page# 34)
- ▶ mathematically
- ▶ precisely
- ▶ accurately

179. The sieve technique works in _____ as follows

- ▶ phases (Page #34)
- ▶ numbers
- ▶ integers
- ▶ routines

180. A (an) _____ is a left-complete binary tree that conforms to the heap order

- ▶ heap (Page# 40)
- ▶ binary tree
- ▶ binary search tree
- ▶ array

181. The sieve technique is a special case, where the number of sub problems is just

- ▶ 5
- ▶ many

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

► 1 (Page #34)

► few

182. Analysis of Selection algorithm ends up with,

► $T(n)$

► $T(1 / 1 + n)$

► $T(n / 2)$

► $T((n / 2) + n)$ (Page #37)

183. For the heap sort we store the tree nodes in

► level-order traversal (Page #40)

► in-order traversal

► pre-order traversal

► post-order traversal

184. The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

► divide-and-conquer (Page# 34)

► decrease and conquer

► greedy nature

► 2-dimension Maxima

185. Theta asymptotic notation for $T(n)$:

► Set of functions described by: $c_1g(n) \leq f(n) \leq c_2g(n)$ for c_1, c_2 s

► Theta for $T(n)$ is actually upper- and worst-case comp

► **Set of functions described by:**

► $c_1g(n)$

186. Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____

► n items (Page# 34)

► phases

► pointers

► constant

187. Memorization is?

► To store previous results for future use

► To avoid these unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later (Page# 47)

► To make the process accurate

► None of the above

188. Quick sort is

► Stable & in place

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

- ▶ Not stable but in place (Page#57)
- ▶ Stable but not in place
- ▶ Sometime stable & some times in place

189. One example of in place but not stable algorithm is

- ▶ Merger Sort
- ▶ Quick Sort (Page# 54)
- ▶ Continuation Sort
- ▶ Bubble Sort

190. Continuation sort is suitable to sort the elements in range 1 to k

- ▶ K is Large
- ▶ K is not known
- ▶ K may be small or large
- ▶ K is small (Page# 57)

191. Which may be a stable sort?

- ▶ Merger
- ▶ Insertion
- ▶ Both above (Page #54)
- ▶ None of the above

192. An in-place sorting algorithm is one that uses ____ arrays for storage

- ▶ Two dimensional arrays
- ▶ More than one array
- ▶ No Additional Array (Page #54)
- ▶ None of the above

193. Continuing sort has time complexity of ?

- ▶ $O(n)$
- ▶ $O(n+k)$
- ▶ $O(n \log n)$
- ▶ $O(k)$

194. single item from a larger set of _____

- ▶ n items (Page# 34)
- ▶ phases
- ▶ pointers
- ▶ v constant

195. For the Sieve Technique we take time

- ▶ $T(nk)$ (Page# 34)
- ▶ $T(n / 3)$
- ▶ n^2

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

► $n/3$

196. One Example of in place but not stable sort is

► Quick (Page# 54)

► Heap

► Merge

► Bubble

197. Consider the following Algorithm:

Factorial (n){

if (n=1)

return 1

else

return (n * Factorial(n-1))

}

Recurrence for the following algorithm is:

☐ ► $T(n) = T(n-1) + 1$

☐ ► $T(n) = nT(n-1) + 1$

☐ ► $T(n) = T(n-1) + n$

☒ ► $T(n) = T(n-1) + 1$

197. Due to left complete nature of binary tree, the heap can be stored in

☐ ► Arrays (Page #40)

☐ ► Structures

☐ ► Link List

☐ ► Stack

198. What type of instructions Random Access Machine (RAM) can execute?

☐ ► Algebraic and logic

☐ ► Geometric and arithmetic

☐ ► Arithmetic and logic (Page# 10)

☐ ► Parallel and recursive

199. What is the total time to heapify?

☐ ► $O(\log n)$ (Page 43)

☐ ► $O(n \log n)$

☐ ► $O(n^2 \log n)$

☐ ► $O(\log^2 n)$

200. word Algorithm comes from the name of the Muslim author _____

☐ ► Abu Jaafar Mohammad ibn Musa al-Khwarizmi.

201. Al-Khwarizmi's work was written in a book titled _____

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

☐ ► al Kitab al-mukhtasar fi hisab al-jabr wa'l-muqabalah

202. Random access machine or RAM is a/an

- ☐ ► Machine build by Al-Khwarizmi
- ☐ ► Mechanical machine
- ☐ ► Electronics machine
- ☐ ► Mathematical model (Page# 10)

203. A RAM is an idealized machine with _____ random-access memory.

- ☐ ► 256MB
- ☐ ► 512MB
- ☐ ► an infinitely large (Page #10)
- ☐ ► 100GB

204. What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?

- ☐ ►
- ☐ ►
- ☐ ► (Page# 14)
- ☐ ►

205. Consider the following code:

```
For(j=1; j<n; j++)
```

```
For(k=1; k<15; k++)
```

```
For(l=5; l<n; l++)
```

```
{
```

```
Do_something_constant();
```

```
}
```

What is the order of execution for this code.

- ☒ ► **$O(n)$**
- ☐ ► $O(n^3)$
- ☐ ► $O(n^2 \log n)$
- ☐ ► $O(n^2)$

206. Is it possible to sort without making comparisons?

- ☐ ► Yes (Page# 57)
- ☐ ► No

207. When we call heapify then at each level the comparison performed takes time

- ☐ ► It will take $\Theta(1)$ (Page# 43)
- ☐ ► Time will vary according to the nature of input data
- ☐ ► It can not be predicted
- ☐ ► It will take $\Theta(\log n)$

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

208. In Quick sort, we don't have the control over the sizes of recursive calls

- ☐ ► True (Page# 40)
- ☐ ► False
- ☐ ► Less information to decide
- ☐ ► Either true or false

209. For Chain Matrix Multiplication we cannot use divide and conquer approach because,

- ☐ ► We do not know the optimum k (Page# 86)
- ☐ ► We use divide and conquer for sorting only
- ☐ ► We can easily perform it in linear time
- ☐ ► Size of data is not given

210. The Knapsack problem belongs to the domain of _____ problems.

- ☐ ► Optimization (Page# 91)
- ☐ ► NP Complete
- ☐ ► Linear Solution
- ☐ ► Sorting

211. Suppose we have three items as shown in the following table, and suppose the capacity of the knapsack is 50 i.e. $W = 50$.

Item Value Weight

- 1 60 10
- 2 100 20
- 3 120 30

The optimal solution is to pick

- ☐ ► Items 1 and 2
- ☐ ► Items 1 and 3
- ☒ ► **Items 2 and 3**
- ☐ ► None of these

212. who invented the quick sort

- ☐ ► C.A.R. Hoare

213. main elements to a divide-and-conquer

- ☐ ► Divide, conquer, combine (Page# 27)

214. Merge sort is a stable algorithm but not an in-place algorithm.

- ☐ ► True (Page# 54)
- ☐ ► false

215. Counting sort the numbers to be sorted are in the range 1 to k where k is small.

- ☐ ► True (Page# 57)
- ☐ ► False

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

216. In selection algorithm, because we eliminate a constant fraction of the array with each phase, we get the
- ☐ ► Convergent geometric series (Page 37)
 - ☐ ► Divergent geometric series
 - ☐ ► None of these
217. In RAM model instructions are executed
- ☐ ► One after another (Page# 10)
 - ☐ ► Parallel
218. Due to left-complete nature of binary tree, heaps can be stored in
- ☐ ► Link list
 - ☐ ► Structure
 - ☐ ► Array (Page# 40)
 - ☐ ► None of above
219. The time assumed for each basic operation to execute on RAM model of computation is-----
- ☐ ► Infinite
 - ☐ ► Continuous
 - ☐ ► Constant (Page# 10)
 - ☐ ► Variable
220. If the indices passed to merge sort algorithm are not equal, the algorithm may return immediately.
- ☐ ► True
 - ☐ ► False (Page# 28)
221. Brute-force algorithm uses no intelligence in pruning out decisions.
- ☐ ► True (Page #18)
 - ☐ ► False
222. In analysis, the Upper Bound means the function grows asymptotically no faster than its largest term.
- ☐ ► True (Page #24)
 - ☐ ► False
223. For small values of n , any algorithm is fast enough. Running time does become an issue when n gets large.
- ☐ ► True (Page #14)
 - ☐ ► False
224. In simple brute-force algorithm, we give no thought to efficiency.
- ☐ ► True (Page# 11)

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

☐ ► False

225. The ancient Roman politicians understood an important principle of good algorithm design that is plan-sweep algorithm.

☐ ► True

☐ ► False (Page# 27)

226. In 2d-space a point is said to be _____ if it is not dominated by any other point in that space.

☐ ► Member

☐ ► Minimal

☐ ► Maximal (Page# 11)

☐ ► Joint

227. An algorithm is a mathematical entity that is dependent on a specific programming language.

☐ ► True

☐ ► False (Page# 7)

228. The running time of an algorithm would not depend upon the optimization by the compiler but that of an implementation of the algorithm would depend on it.

☐ ► True (Page# 13)

☐ ► False

229. $F(n)$ and $g(n)$ are asymptotically equivalent. This means that they have essentially the same _____ for large n .

☐ ► Results

☐ ► Variables

☐ ► Size

☐ ► Growth rates (Page#23)

230. $8n^2 + 2n - 3$ will eventually exceed $c2^*(n)$ no matter how large we make $c2$.

☐ ► True (Page# 25)

☐ ► False

231. If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a _____ car.

☐ ► Fast

☐ ► Slow

☐ ► Expensive

☐ ► Cheap (Page# 11)

232. The function $f(n) = n(\log n + 1)/2$ is asymptotically equivalent to $n \log n$. Here Upper Bound means the function $f(n)$ grows asymptotically _____ faster than $n \log n$.

☐

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

► More

► Quiet

► Not (Page# 24)

► At least

233. Counting Money problem is an example which cannot be optimally solved by greedy algorithm.

☐ ► True (Page# 97)

☐ ► False

234. Huffman algorithm generates an optimum prefix code.

☐ ► True (Page #102)

☐ ► False

235. If the string "lmncde" is coded with ASCII code, the message length would be _____ bits.

☐ ► 24

☐ ► 36

☒ ► 48 ($6 \times 8 = 48$)

☐ ► 60

236. There are _____ nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.

☐ ► 2

☐ ► 3 (Page# 90)

☐ ► 4

☐ ► 5

237. Inductive approach to compute entries of V is implied in _____ based solution of knapsack problem.

☐ ► Brute force

☐ ► Dynamic programming (Page #93)

238. A number of lectures are to be given in a single lecture hall. Optimum scheduling for this is an example of Activity selection.

☐ ► True (Page# 105)

☐ ► False

239. The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an _____ solution.

☐ ► Simple

☐ ► Sub optimal

☐ ► Optimal (Page# 105)

☐ ► Non optimal

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

240. The string |xyz|, if coded with ASCII code, the message length would be 24 bits.

☐ ► True ($3 \times 8 = 24$)

☐ ► False

241. An application problem is one in which you want to find, not just a solution, but the ____ solution.

☐ ► Simple

☐ ► Good (Page# 113)

☐ ► Best

☐ ► Worst

• A free tree with n vertices has exactly ____ edges.

n

$n+1$

c. $n-1$ page 142

1

3) Kruskal's algorithm works by adding ____ in increasing order of weight

(lightest edge first)

• Verticals

b. Edges page 147

• Trees

• Weights

6) Computing the strongly connected components of a digraphs is a/an ____ of the

problem to determine whether a diagram is strongly connected or not

• Size

b. Generalization page 135

10) The process of updating estimates in Dijkstra's algorithm is called ____

• Updating

• Amendment

c. Relaxation page 154

• Insertion

• An un-weighted graph can be considered as a graph in which every edge has weight ____ Unit.

7

5

3

d. 1 page 153

14) The breadth-first-search algorithm is a shortest-path algorithm that works

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

on _____

graphs.

- Weighted

- Directed

c. Un-weighted page 153

d. Un-directed

15) Which activity creates a unique cycle in a free tree:

- adding any vertex

- adding any sub tree

- adding root

d. adding any edge page 142

8. An edge (u, v) _____ $E - A$ is safe if A _____ $\{(u, v)\}$ is viable.

a. $\mathcal{E}, U$ page 143

- $U, \mathcal{E}$

- $\mathcal{E}, \Pi$

- Π, U

18) The relationship between number of back edges and Number of cycles in DFS is,

- Both are equal

- Back edges are half of cycles

- Back edges are one quarter of cycles

d. There is no relationship between no. of edges and cycles google

10. From given algorithms which one considered as best for finding the shortest-path:

19) DFS

20) Bellman-Ford algorithm

c. Dijkstra's algorithm page 154 / google

d. BFS

11. Overall time for Kruskal algorithm is

a. $\Theta(\log E)$

b. $\Theta(E \log V)$ page 149 / google

12. Dijkstra's Algorithm is used to solve _____ problems.

- a. All-pair shortest path

- b. Single-source shortest path page 154

- Multi-source shortest path

- Sorting & searching

13. A digraph is strongly connected under what condition?

22) A digraph is strongly connected if for every pair of vertices $u, v \in V$, u can

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

reach v.

23) A digraph is strongly connected if for every pair of vertices $u, v \in V$, u can

reach v and vice versa. Page 135

22) A digraph is strongly connected if for at least one pair of vertex $u, v \in V$, u can reach v and vice versa.

23) A digraph is strongly connected if at least one third pair of vertices $u, v \in V$, u can reach v and vice versa.

14. Floyd-Warshall Algorithm is based on _____

a. Dynamic Programming page 161

- Greedy Approach
- Divide and Conquer
- Complexity theory

15. In strong components algorithm, vertices are sorted in _____ order of finish times.

- a. Any
- b. Increasing
- c. Decreasing page 141 / google
- d. Strong

16. _____ is commonly the running time of Dijkstra's Algorithm using the binary heap method.

- a. $\Theta(E \log V)$ page 156
- $\Theta(V \log V)$
- $\Theta(\log E)$
- $\Theta(\log V)$

17. Which technique is used in the implementation of Kruskal solution for the MST?

a. Greedy Technique page 142 / google

26) Divide-and-Conquer Technique

27) Dynamic Programming Technique

28) The algorithm combines more than one of the above techniques i.e. Divide-and-Conquer and Dynamic Programming

18. A fully connected undirected graph of 5 nodes will have _____ edges.

28) 4

29) 5

c. $10 \cdot n(n-1)/2$

d. 15

19. Prim's algorithm is based on _____ strategy.

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTACT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

- ★ a. Greedy page 150 / google
- ★ b. Dynamic programming
- ★ c. Divide and Conquer
- ★ d. Exponential
- ★ 20. We say that two vertices u and v are mutually _____ if u can reach v and vice versa.
- ★ 29) Crossed
- ★ 30) Forward
- ★ c. Reachable page 135
- ★ d. Not Reachable
- ★ 21. In Dijkstra's algorithm, initially the estimated value from source vertex to any vertex v is:
 - ★ a. Zero (0)
 - ★ b. One (1)
 - ★ c. Minus one (-1)
 - ★ d. Infinity (∞) page 154
- ★ 22. A strongly connected component only apply to:
 - ★ a. Directed Graph page 135 / google
 - ★ • Undirected Graph
 - ★ • Minimum Spanning Tree
 - ★ • Breadth First Search
- ★ 23. A graph may contain _____
 - ★ 32) Exactly one MST
 - ★ 33) No MST
 - ★ 34) One or zero MST
 - ★ d. More than one MST google
- ★ 24. In _____ algorithm(s), at any time, the subset of edges A forms a single tree.
 - ★ a. Kruskals
 - ★ b. Prim's page 149
 - ★ 32) kruskal's and Prim's
 - ★ 33) BFS
- ★ 25. The _____ given by DFS allow us to determine a number of things about a graph or digraph
 - ★ a. color stamps
 - ★ b. time stamps page 130
 - ★ c. line stamps
 - ★ d. node stamps

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

26. In Kruskal's algorithm, the vertices will be stored in _____.
a. links
b. sets page 147
• nodes
• Loops
27. Keeping in mind the shortest path, if given scenarios occur in computer networks like the internet where data packets have to be routed. The vertices are _____. Edges are _____ which may be wired or wireless.
a. Routers, communication links page 153
• Internet, routers
• Communication links, routers
• Routers, internet
28. In computing the strongly connected components of a digraph. vertices of the digraph are _____ into subsets
a. Joined
b. Partitioned page 135
c. Deleted
d. Created
29. In undirected graph, by convention all the edges are called _____ edges.
a. Forward
b. Back page 130
37) Cross
38) Both forward and back
30. In Timestamped DFS, No back edges means _____
a. 1 cycle
b. no cycles page 131
38) DFS
39) BFS
31. In Prim's algorithm, we start with the _____ vertex r, it can be any vertex.
• Pivot
• Leaf
• negative
d. Root page 149
32. The ancestor and descendent relation can be nicely inferred by the _____ lemma.
43) addition
44) division

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

- ★ c. parenthesis page 129
- ★ d. node

★ 33. Which of the following statement is true?

- ★ • Kruskal algorithm is multiple source technique for finding MST.
- ★ • Kruskal's algorithm is used to find minimum spanning tree of a graph, time complexity of this algorithm is $O(EV)$
- ★ • Both I and II
- ★ • Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best Tree edge) when the graph has relatively few edges.
- ★ See option B of Sr 46

★ 34. The time complexity to compute Graph transposes G^T is $(V+E)$, if you have ___ for G .

- ★ a. an adjacency list page 138
- ★ • Array list
- ★ • complete list
- ★ • stack

★ 35. In Prim's algorithm, If there is no edge from u to a vertex in S . we set the key value to _____

- ★ • 0
- ★ • 1
- ★ • -1
- ★ d. ∞ page 151

★ 36. What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?

- ★ • $O(\log E)$
- ★ • $O(V)$
- ★ • $O(V+E)$
- ★ d. $O(\log V)$ page 152

★ 37. Edge weights can be interpreted as distance _____

- ★ • in breadth-First Search
- ★ • in Queue's
- ★ c. in the shortest-paths page 153
- ★ d. in depth-First Search

★ 38. The _____ given by DFS allow us to determine whether the graph contains any cycles.

- ★ a. Order
- ★ b. Time stamps page 130

★ 47) BFS traversing

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

- 48) Topological sort
39. Dijkstra's algorithm works on a weighted directed graph $G(V, E)$ in which all weights _____ are non-negative.
- a. vertices
 - b. edges page 154
 - nodes
 - links
40. By breaking any edge on a cycle created in free tree, the free _____ is restored.
- a. Edge
 - b. Tree page 142
- 49) Cycle
- 50) Vertex
41. Bellman-Ford algorithm is used to solve _____ problems.
- a. All pair shortest path
 - b. Single source shortest path google
- 49) Flow of networking
- 50) Double source shortest path
42. From the given options which one is correct regarding the time complexity of Dijkstra's algorithm
- $O(N)$
 - $O(N^3)$
 - $O(N^2)$
 - d. $O(\log N)$ google
43. In which algorithm, information of shortest path is propagated sequentially along each shortest path in the graph.
- a. Bellman-Ford page 16
- 52) Brute-force technique
- 53) Dijkstra's
- 54) Prim's
44. In Timestamped DFS-cycles lemma, if edge (u, v) is a tree, forward or cross edge, then
- a. $f[u] < f[v]$
 - b. $f[u] > f[v]$ page 130
- 52) $f[u] \leq f[v]$

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

53) $f[u] \geq f[v]$

45. Bellman-Ford Algorithm does not allow $G(\text{graph})$ to have _____

- a. positive cost cycles
- b. negative cost cycles page 159

53) negative weights edges

54) positive weights edges

46. Kruskal algorithm (choose best non cycle edge) is better than Prim's (choose best tree edge) when the _____ has relatively few _____

- a. tree, edges
- b. graph edges see option D of Sr 23
- tree, branches
- graph, branches

47. Finding the faster result of the shortest path from u to v for every pair of vertices and we use _____

54) Single-pair shortest-paths problem

55) Two pair shortest the problem

c. All pair shortest paths problem page 153

d. both I and II

48. There are no _____ edges in undirected graph.

- a. Forward
- b. Back
- c. Cross page 130
- d. Both forward and back

49. Networks are _____ in the sense that it is possible from any location in the network to reach any other location in the digraph

a. Complete page 135

56) Incomplete

57) Not graphs

58) Transportation

50. Dijkstra's algorithm:

- a. Has greedy approach to find all shortest paths
- b. Has both greedy and Dynamic approach to find all shortest paths
- c. Has greedy approach to compute single source shortest paths to all other vertices page 154

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

- 57) Has both greedy and dynamic approach to compute single source shortest paths to all other vertices
51. Bellman-Ford algorithm is slower than
- a. Brute-force technique
 - b. Dijkstra's page 159
- 58) Prim's
- 59) Graph Algorithm
52. Dijkstra's Algorithm cannot be applied on
- a. directed and weighted graphs
 - b. graphs having negative weight function (google)
53. Which of the following is used in the data structure for implementing Dijkstra's Algorithm?
- 59) Max heap
 - 60) Stack's
 - 61) Circular queue
 - d. Priority queue (Google)
54. In Generic approach determining of Greedy MST, we maintain a subset A of ____
- a. Edges page 143
 - Vertices
 - Cycles
 - Paths
55. In Dijkstra's algorithm the estimated value of source vertex $d[s]$ is
- a. Equal to 0 page 154
 - Equal to 1
 - Greater than 0
 - Greater than 1
56. Dijkstra's algorithm is a simple _____ algorithm for computing the single-source shortest-paths to all other vertices.
- a. Greedy page 154
- 62) Bellman-Ford
- 63) Divide and conquer
- 64) Brute-Force

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

57. In Timestamped DFS. If there is a back edge (u, v) then v is an ancestor of u and by following tree edge from v to u , we get _____.

- a. nothing
- b. a cycle page 131
- 62) a line
- 63) a graph

58. There exists a unique path between any _____ vertices of a free tree.

- a. One
- b. Two page 142
- Three
- Four

59. _____ technique is look like propagating wave-front outward.

- a. Generic traversal
- b. Breadth first traversal page 117
- 64) Depth first traversal

60. You have an adjacency list for G , what is the time complexity to compute

Graph transpose G^T ?

- a. $(V+E)$ page 138
- $V.E$
- V
- E

61. In the shortest-paths problem, we are given a weighted of _____ $G=(V, E)$.

- a. Directed graph page 153

- 65) Line graph
- 66) Un-directed graph
- 67) Weighted graph

62. Equivalence relation partitions the vertices into _____ classes of mutually reachable vertices and these are the strong components

- a. Variance
- b. Equivalence page 136
- 66) Non equivalence
- 67) Non classes

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

63. Overall Running time of Prim's algorithm is _____.
67) $\Theta(\text{ElogE})$
68) $\Theta(\text{ElogV})$
c. $\Theta((V+E)\log V)$ page 152
d. $\Theta((V+E)\log E)$
64. For _____ graphs, there is no distinction between forward and back edges.
a. large
b. directed
c. undirected page 130
d. medium
65. For _____ graphs, there is no distinction between forward and back edges.
a. Undirected page 130
• directed
• small
• large
66. As the Kruskal's algorithm runs, the edges in viable set A induce a _____ on the vertices.
a. Set
b. Graph
c. Tree
d. Forest page 147
67. In strong components algorithm, the form of graph is used in which all the _____ of original graph G have been reversed in direction.
a. Vertices
b. Edges page 138
c. Both edges & vertices
d. Trees
68. In computing the _____ components of a digraph, vertices of the digraph are partitioned into subsets.
a. weakly connected
b. strongly connected page 135
c. best
d. worst
69. Timestamp structure of _____ is used in determining the strong components of a digraph.
a. DFS google
b. BFS

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

- c. Both DFS & BFS
- d. MST

70. In Prim's algorithm, if the color of a vertex is _____ then it is in S otherwise not.

- a. White
- b. Gray
- c. Black page 151
- d. Blue

71. Digraphs _____ in communication and transportation networks.

- a. are not used
- b. are used page 135
- c. parts are used
- d. final value is used

72. Once you enter a strong component, every vertex in the component is _____.

- a. not reachable
- b. reachable page 137
- c. reachable some times
- d. removed

73. In Kruskal's algorithm, the next edge is added to viable set A, if its adding does not induce a/an _____.

- a. Vertex
- b. Edge
- c. Cycle page 147
- d. Tree

74. Problems such as the shortest route between cities can be solved efficiently by modelling the road map as a _____.

- a. Tree
- b. Graph page 153
- c. Linked list
- d. Stack

75. In Bellman-Ford Algorithm, relaxation applies to every edge of the graph and repeat this _____ time.

- a. $E - 1$
- b. $E + 1$
- c. $V + 1$
- d. $V - 1$ page 159

76. A topological sort of a DAG is a _____ ordering of the vertices of the DAG such

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

that for each edge (u, v) , u appears before v in the ordering.

- a. Linear page 134
- b. Parallel
- c. Sequence
- d. Non-linear

77. Adding any edge to a free tree creates a unique _____

- a. Vertex
- b. Edge
- c. Cycle page 142
- d. Strong component

78. Back edge is:

- a. (u, v) where v is an ancestor of u in the tree. Page 128
- b. (u, v) where u is an ancestor of v in the tree.
- c. (u, v) where v is a predecessor of u in the tree.
- d. (u, v) where u is a mid of v in the tree.

79. In Kruskal's algorithm, the next _____ is not added to viable set A , if its adding induce a/an cycle.

- a. Vertex
- b. Edge page 147
- c. Cycle
- d. Tree

80. Which of the following statement is false about Dijkstra's Algorithm?

- a. It can be applied on graphs having a negative weight function google
- b. It is used to solve Single-source shortest path
- c. It works on a weighted directed graph
- d. Its implementation in data structure is possible through the priority queue 81. In

strong components algorithm, first of all DFS is run for getting _____ times of vertices.

- a. Start page 138
- b. Finish
- c. Both start & finish
- d. Middle

82. The tricky part of _____ algorithm(s) is/are, how to detect whether the addition of an edge will create a cycle in viable set A .

- a. Kruskal's page 149
- b. Prim's
- c. Both Kruskal's and Prim's
- d. DFS

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

83. _____ components are not affected by reversal of all edges in terms of vertices reachability.

- a. Strongly connected page 139
- b. Weakly connected
- c. First two
- d. Last two

84. In Prim's algorithm, we will make use of _____.

- a. Stack
- b. Priority Queue page 150
- c. Array
- d. List

85. The component digraph is necessarily _____.

- a. straight
- b. cyclic
- c. acyclic page 136
- d. strong

86. A free tree with n _____ have exactly $n - 1$ _____.

- a. vertices, nodes
- b. edges, vertices
- c. nodes, vertices
- d. vertices, edges page 142

87. For each vertex $u \in (V-S)$, we associated Key

- a. Key[v]
- b. Key[s]
- c. Key[u] page 151
- d. Key[v-s]

88. If you find yourself in maze the better traversal approach will be:

- a. BFS google (always return with short part)
- c. Level order
- d. DFS

89. Forward edge is :

- a. (u, v) where u is proper descendent of v in the tree
- b. (u, v) where v is proper descendent of u in the tree page 129
- c. (u, v) where v is proper ancestor of u in the tree
- d. (u, v) where u is proper ancestor of v in the tree

SANAULLAH KHAN
MSc. MATHEMATICS
FOR QUIZ CONTECT
(03066793776)

YOUTUBE CHANNEL: S KHAN ACADEMY

90. cross edge is :
- a. (u, v) where u and v are not ancestor of one another
 - b. (u, v) where u is ancestor of v and v is not descendent of u
 - c. (u, v) where u and v are not ancestor or descendent of one another page 129
 - d. (u, v) where u and v are either ancestor or descendent of one another
91. The running time of Bellman-Ford algorithm is _____.
- a. $\Theta(V + E)$
 - b. $\Theta(E + E)$
 - c. $\Theta(VE)$ page 159
 - d. $\Theta(V + V)$
92. In Bellman-Ford Algorithm, relaxation applies to _____ of the graph.
- a. Every edge page 159
93. If a subset of edges A is visible for building MST, it cannot contain a/an _____.
- a. Vertex
 - b. Edge
 - c. Cycle page 143
 - d. Graph
94. In Timestamped DFS-cycles lemma, if edge (u, v) is a back edge, then _____.
- a. $f[u] < f[v]$
 - b. $f[u] > f[v]$
 - c. $f[u] \leq f[v]$ page 130
 - d. $f[u] \geq f[v]$
95. In digraph $G=(V,E)$; G has cycle if and only if
- a. The DFS forest has forward edge.
 - b. The DFS forest has back edge page 131
 - c. The DFS forest has both back and forward edge
 - d. BFS forest has forward edge
96. The key $[u]$ is the weight of the _____ going from u to any vertex in S .
- a. lightest edge page 151
97. In Bellman-Ford Algorithm, path consists of at most _____ edges.
- a. $V + 1$
 - b. $V-1$ page 160
 - c. $E + 1$
 - d. $E-1$
98. According to parenthesis lemma, vertex u is a descendent of v vertex it and only if;

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YOUTUBE CHANNEL: S KHAN ACADEMY

- ★ a. $[d[u], f[u]] \subseteq [d[v], f[v]]$ page 129
- ★ b. $[d[u], f[u]] \supseteq [d[v], f[v]]$
- ★ c. Unrelated
- ★ d. Disjoint

- ★ 99. According to parenthesis lemma, vertex u is an ancestor of v vertex if and only if; a. $[d[u], f[u]] \subseteq [d[v], f[v]]$
- ★ b. $[d[u], f[u]] \supseteq [d[v], f[v]]$ page 129
- ★ c. Unrelated
- ★ d. Disjoint

- ★ 100. Edge weights can be interpreted as distance ____.
 - ★ a. in breadth-First Search
 - ★ b. in Queue's
 - ★ c. in the shortest-paths page 153
 - ★ d. in depth-First Search
- ★ 101. _____ algorithm allows negative weights edges and no negative cost cycles.
 - ★ a. Brute-force technique
 - ★ b. Bellman-Ford page 159
 - ★ c. Dijkstra's
 - ★ d. Prim's

- ★ In Huffman encoding, the characters with smallest probabilities are placed at the-----depth of the tree.
- ★ • Minimum
- ★ • Average
- ★ • Maximum page 102
- ★ • Root

- ★ There are ----- ways to representing graphs.
- ★ • 3
- ★ • 1
- ★ • 2 page 116
- ★ • 4

- ★ Each time we traverse graph by Breadth-first search algorithm, we count the distance from---
- ★ -----.
- ★ • Starting node
- ★ • Neighbors of the starting node page 117
- ★ • Right most node
- ★ • Left most node

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In Activity Selection problem, intuitively-----.

- There are always short activities as input
- Short activities are not attractive
- Duration of the activities does not matter
- We do not like long activities page 105

65) The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an-----solution.

- Simple
- Sub-optimal
- Optimal page 105
- Non optimal

The ----- is a problem for which the greedy algorithm approach provides an optimal solution.

- Activity selection page 105
- Dynamic programming
- Knapsack problem
- NP complete problem

Which of the following ways can be used to represent a graph?

- Adjacency list
- Incidence matrix
- Adjacency list , Adjacency Matrix page 116
- No way to represent

Queue Data Structure work on ----- principles.

- FIFO(first in first out) cs301

Graphs can be represented by an----- and -----.

- queue , stack
- adjacency list , adjacency matrix page 116
- adjacency right , adjacency left
- Binary , linear

Identify a TRUE statement about Knapsack.

- The Knapsack Problem does not belongs to the domain of optimization problems
- The Knapsack Problem belongs to the domain of optimization problems page

The running time of brute-force algorithm to solve Knapsack problem is-----

- $O(n)$

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- $O(n^3)$
- $O(n!)$
- $O(2^n)$ pg#92

If a matrix has three rows and two columns, the dimension of matrix will be:

- 3×2 Conceptual
- 2×3
- 3×3
- 2×2

The Probability in Huffman encoding is the number of occurrences of a character divided by the total-----in the message.

- Numbers
- Frequencies
- Strings
- Characters pg#100

In recursive formulation of Knapsack

Problem: $V[0, j] = \text{-----}$ for $j \geq 0$

- -1
- 0 pg#93
- 1
- 2

The Knapsack Problem belongs to the domain of-----problem.

- Optimization pg#91
- Sorting
- Linear solution
- Searching

An optimization problem is one in which you want to find the-----solution.

- Simple
- Good
- Best pg#97
- Worst

Huffman algorithm generates an optimum----- code.

- Prefix page 102
- Postfix
- Infix
- None of the above

Which of the following algorithms solves the Fractional Knapsack Problem more effectively?

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- Divide and Conquer
- Greedy algorithm page 109

If we implement the bag by using a queue, we have-----.

- BFS page 124
- DFS
- Graph
- Loop

One of the limitation in 0/1 Knapsack is that an item can either be-----in the bag or not.

- Use
- Put page 91
- Move
- Store

In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object 'i' then we gain-----.

- W(Total weight of Knapsack)
- V(Total value of all items)
- v_i (value of object i) page 93
- None of the given option

An activity scheduling algorithm, the width of a rectangle-----.

- Indicates the duration of an activity page 106

The prefix code generated by Huffman algorithm----- the expected length of the encoded string.

- Minimizes page 102
- Balances
- Maximizes
- Keeps constant

WORK.100%

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